State-space Gromov–Hausdorff convergence of spectral truncations of compact semisimple Lie groups with bi-invariant metric

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Abstract

Let G be a compact connected semisimple Lie group of rank $r \geq 1$ equipped with a bi-invariant Riemannian metric, and let $\mathcal{T}_G = (C^\infty(G), L^2(G,\Sigma), D_G)$ denote its canonical metric spectral triple, with D_G the Kostant cubic Dirac operator. For each cutoff $\Lambda > 0$, let $\mathcal{T}_\Lambda = (\mathcal{O}_\Lambda, \mathcal{H}_\Lambda, P_\Lambda D_G P_\Lambda)$ denote the Connes–van Suijlekom spectral truncation obtained by projection onto irreducibles with Casimir $\leq \Lambda^2$. We prove that (for SU(2); for general G conditionally on the per-group spin-window decomposition, see below) $\mathcal{T}_\Lambda$ converges to $\mathcal{T}_G$ in van Suijlekom's state-space Gromov–Hausdorff distance, with the state spaces metrized by the left-translation Lipschitz seminorm as in Paper 38 (convergence in the Dirac-commutator seminorm is open), with

$$d_{\mathrm{GH}}(\mathcal{T}_\Lambda, \mathcal{T}_G) \;\leq\; \gamma_\Lambda(G) \;\to\; 0, \qquad \gamma_\Lambda(\mathrm{SU}(2)) = \frac{2}{\pi}\,\frac{\log\Lambda}{\Lambda} + O(1/\Lambda),$$

with constant 1 from the lifted-state route of Paper 38 (the Lipschitz comparison constant $C_3(G) = 1$ of Lemma 3.11 enters only the withdrawn approximation-pair route). The SU(2) moment constant $2/\pi$ **[INTERNAL THEOREM]** is Paper 38's $(4/\pi)\log n/n$ read in the cutoff variable used here: $\mathrm{Cas}(\mathrm{ad}) = h^\vee = 2$ on $\mathfrak{su}(2)$ fixes $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$, whose geodesic distance from the identity is the rotation angle — exactly the distance Paper 38 integrates against — and Paper 38's shell index is $n_{\max} = 2j_{\max} + 1$, so $n \simeq 2\Lambda$. It is a kernel constant, not the convergence rate: a tapered amplitude gives $O(1/\Lambda)$ on SU(2) **[MEASURED]**, Paper 38; for general G this is open. Because $\gamma_\Lambda\Lambda$ is invariant under rescaling the bi-invariant metric, the constant in Λ does not depend on the normalisation; Paper 38's values $4/\pi$ (dual-Coxeter) and $2/\pi$ (unit S^3) are per its shell index n. Whether the leading constant $c(G)$ of $\gamma_\Lambda(G)$ is the same for every compact connected simple G is **[OPEN]**. Paper 38 [26] is recovered as the rank-1 case $G = \mathrm{SU}(2)$. The proof is the lifted-state route (UCP compression plus the conjugation average against the L2 central spectral Fejér kernel with closed-form quantitative rate); the lemma chain (L1′ operator system, L3 Lipschitz comparison with constant 1 at every rank, proved by compression, L4 Berezin reconstruction via Peter–Weyl) supplies the substrate and supporting estimates, L3 entering only the withdrawn approximation-pair route. The approximation pair of §3.5 is recorded but *does not supply an independent proof*: its dual reach is a named gap and both height constituents are refuted (Lemma 3.28, corrected 2026-09-04). **[OBSERVATION]** Paper 38 reads its Fejér-moment constant (a reading, not a derivation) as the $k = 0$ volume-slot content of a master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$ on the spectral triple; whether the value so read extends from SU(2) to the other simple groups is open with $c(G)$: a Plancherel × Vandermonde cancellation is proposed as the mechanism (a proof sketch), and the rank-2 and rank-3 extractions reported here compare mismatched conventions and do not test it (status note in §3.2). For SU(2) the result supplies an explicit rate (translation-seminorm metrization) for

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the state-space Gromov–Hausdorff convergence proved by Gaudillot-Estrada and van Suijlekom [10]; for general G it gives the bound $d_{\mathrm{GH}} \leq \gamma_\Lambda(G) \to 0$, conditionally, with the rate form open. The strictly stronger Latrémolière propinquity is *not* claimed: its dual-reach step is a named gap (Remark 3.5), exactly as for the rank-1 case (Paper 38). Moreover the general-G lifted-state construction underlying even the state-space GH statement is verified only for SU(2), so for general G the convergence is conditional on that per-group spin-window decomposition; the closed-form moment $\gamma_\Lambda(G)$ (the genuine new content) is unaffected. For semisimple G with more than one simple factor the leading constant is open (Remark 3.9).

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1 Introduction

The convergence of finite-dimensional truncations of a Riemannian spectral triple to its continuum limit, in a suitable quantum Gromov–Hausdorff sense, is the central metric problem in noncommutative geometry as developed by Connes–van Suijlekom [7] and Latrémolière [18]. The Connes–van Suijlekom truncated operator system $\mathcal{O}_\Lambda = P_\Lambda \mathcal{A} P_\Lambda$ encodes a finite-dimensional spectral fragment of the continuum algebra obtained by projection onto the modes of the Dirac operator with $|D| \leq \Lambda$. The natural question is whether the sequence $(\mathcal{O}_\Lambda)_\Lambda$ converges, as a sequence of metric spectral triples, to the continuum geometry, and at what rate.

The flat-structure case was worked out in several papers between 2022 and 2024. Hekkelman [12] treated the circle S^1 using Toeplitz/Fejér reconstruction. Leimbach and van Suijlekom [21] extended the analysis to flat tori T^d, proving state-space Gromov–Hausdorff convergence of truncated tori (they state no explicit rate; an earlier version of this paper attributed an $O(1/\Lambda)$ rate to them, withdrawn 2026-09-28), using the standard Schur multiplier characterisation of the abelian-compact-group Fejér mechanism. A complementary Kähler line of work [11, 23] treats Berezin–Toeplitz reconstruction on S^2 (matrix algebras converging to the coadjoint orbit), using the Kähler structure of the coadjoint orbit. All of these results address flat abelian or Kähler structures.

The scalar compact-group case is due to Gaudillot-Estrada–van Suijlekom [10]; the present author's Paper 38 [26] treats the spinor (Camporesi–Higuchi) instance on the round $S^3 = \mathrm{SU}(2)$ Camporesi–Higuchi spectral triple with explicit rate $d_{\mathrm{GH}}(\mathcal{T}_\Lambda, \mathcal{T}_{S^3}) \leq \gamma_\Lambda = O(\log\Lambda/\Lambda)$ (a tapered amplitude gives $O(1/\Lambda)$ **[MEASURED]**), the Fejér moment having asymptotic constant $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ in Paper 38's shell index n ($2/\pi$ in the cutoff Λ used here, $n \simeq 2\Lambda$), twice the quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 2/\pi$ of unit-radius sphere volumes (not the Hopf fibration's base-to-total ratio; see Paper 38's volume-ratio note). The proof proceeded through a lemma chain L1′–L4 plus an assembly step (the fifth, §3.5, has since been reduced to a record rather than a proof — dual reach a named gap, both heights refuted), in which the principal new ingredients beyond the abelian case were (i) a central spectral Fejér kernel on SU(2) built from Peter–Weyl rather than Schur–Fourier multipliers, (ii) a Lipschitz comparison estimate (originally argued through the SO(4) selection rule on Avery 3-Y integrals; since 2026-09-28 proved by compression, Paper 38 Lemma L3), and (iii) a non-Kähler Berezin reconstruction via the central-Fejér-kernel spectral form.

Concurrent state-space result. While Paper 38 was in preparation, Gaudillot-Estrada and van Suijlekom [10] proved, in a structurally parallel direction, that the Peter–Weyl truncated

state spaces of *any* compact metric group with bi-invariant metric converge in the classical (not quantum) Gromov–Hausdorff distance to the full state space. Their result covers $\mathrm{SU}(N)$, $\mathrm{Spin}(n)$, $\mathrm{Sp}(n)$, and the exceptional compact Lie groups in a single statement, but *does not involve the Dirac operator* and *does not give an explicit rate*. They observe explicitly that quantum and classical Gromov–Hausdorff distance may not agree, leaving the spectral-triple-level propinquity question open at the class level.[1]

What this paper does. We extend Paper 38 toward [10] on the semisimple sub-class of Class 1 (compact connected semisimple Lie groups; the leading moment constant is derived for SU(2) and open for the other simple groups, and a central torus factor, which carries the classical circle Fejér constant, is excluded, see Remark 3.9).

Theorem 1.1 (Main theorem; cf. §4)**.** *Let G be a compact connected semisimple Lie group of rank $r \geq 1$ with bi-invariant Riemannian metric (dual-Coxeter normalised). The truncated metric spectral triples $\mathcal{T}_\Lambda$ converge to $\mathcal{T}_G$ in van Suijlekom's state-space Gromov–Hausdorff distance, with the state spaces metrized by the left-translation Lipschitz seminorm (convergence in the Dirac-commutator seminorm is open), with explicit rate*

$$d_{\mathrm{GH}}(\mathcal{T}_\Lambda, \mathcal{T}_G) \;\leq\; \gamma_\Lambda(G) \;\to\; 0, \qquad \gamma_\Lambda(\mathrm{SU}(2)) = \frac{2}{\pi}\,\frac{\log\Lambda}{\Lambda} + O\big(1/\Lambda\big), \tag{1}$$

with constant 1 *from the lifted-state route of Paper 38 (the Lipschitz comparison constant $C_3(G) = 1$ of Lemma 3.11 enters only the withdrawn approximation-pair route), where $\gamma_\Lambda(G)$ is the mass-concentration moment of the central spectral Fejér kernel on G given by Lemma 3.4. The displayed asymptotic is the* SU(2) *case* ***[INTERNAL THEOREM]*** *(a kernel constant; whether a tapered amplitude improves the rate for general G, as it does on* SU(2) ***[MEASURED]****, is open). For the other simple G the bound and $\gamma_\Lambda(G) \to 0$ hold, and whether the leading constant is the same is* ***[OPEN]****; for several simple factors only the bound and $\gamma_\Lambda(G) \to 0$ survive (Remark 3.9).* Scope of the hypothesis: *the statement is unconditional for $G = \mathrm{SU}(2)$; for general G it is conditional on the per-group spin-window decomposition underlying the lifted-state construction (Remark 3.5), verified only in the rank-one case. The closed-form moment — the new content here — does not depend on that hypothesis. The stronger Latrémolière propinquity is not claimed at any rank (Remark 3.5).*

Whether the leading constant $c(G)$ of $\gamma_\Lambda(G)$ is the same for every compact connected simple G is **[OPEN]**. Two natural rank-r generalisations were proposed in the supporting internal scoping memo, both normalised to Paper 38's per-n value $4/\pi$: a *group-independent* reading (Reading A, $c(G) = c(\mathrm{SU}(2))$), and a *Weyl-formula* reading (Reading B, $c(G) = 2|W(G)|/\pi^r$) that would imply substantial rank-dependence. We extracted the constant numerically on SU(3), $\mathrm{Sp}(2) = \mathrm{Sp}(4)$, G_2 and SU(4). The extraction pipeline divides by $\sqrt{C_{\mathrm{gen}}(\mathrm{ad})/h^\vee}$ while leaving the distances unscaled; applied to SU(2) it returns $(2/\pi)/\sqrt{2} \approx 0.450$, not the anchor $2/\pi$. The proximity of its rank-2 outputs to $4/\pi$ therefore compares mismatched conventions, and the extraction discriminates neither for nor against Reading A, at G_2 included (Remark 3.7).

At rank 1 the constant is derived (**[INTERNAL THEOREM]**; SU(2): Paper 38 [26], with the closed-form sum-rule of §3.2 verified to $4/\pi$ per n, i.e. $2/\pi$ per Λ, by the doubling estimator in `geovac/central_fejer_su2.py`). For general G the Plancherel-weight × Vandermonde-Jacobian cancellation (Remark 3.6) proposes a *mechanism* as a proof sketch (§3.2); a fully rank-uniform analytical proof is a named gap. In parallel, the Lipschitz comparison constant $C_3(G) = 1$ holds at every rank and every cutoff by compression (Lemma 3.11, 2026-10-02). The earlier per-summand analysis (the Dirac-triangle and intermediate inequalities, via the

[1] Companion work in the quantum-group setting by Leimbach [20] extends the state-space result to coamenable compact quantum groups including $\mathrm{SU}_q(N)$, again with no Dirac and no rate.

Parthasarathy–Ranga Rao–Varadarajan summand bound and a Brauer–Klimyk signed-sum closure, case-bookkeeping verified at 5641/5641 cells across $\mathrm{SU}(3), \mathrm{SU}(4), \mathrm{Sp}(2), G_2$) is retained as representation-theoretic statements about Kostant eigenvalues and Casimirs (proved for the trivial-λ and Cartan-product summands, verified on weight-bounded panels for the PRV and interior classes); its bridge to the commutator norm (an entrywise bound read as an operator-norm bound, and eigenvalue ratios read as per-isotypic Lipschitz constants) is the step Paper 38 refuted at rank 1 on 2026-09-28 and is withdrawn here, so it is neither needed for nor a check of Lemma 3.11.

Structural reading. **[OBSERVATION]** Paper 38's per-n constant $4/\pi$ is read (a reading, not a derivation; Paper 38) as the *master Mellin M1 volume-slot* content on the spectral triple, in the sense that $4/\pi = 2\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)) = \mathrm{Vol}(S^2)/\pi^2$ is the volume quotient $2\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ at unit radius (Paper 38, volume-ratio note), and is read as the value of the $k = 0$ slice of the spectral-action heat-kernel Mellin transform $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$ on the canonical Riemannian spectral triple. Whether the value so read is the same across the simple groups is open with $c(G)$; if the proposed cancellation mechanism holds, the central spectral Fejér kernel on any simple group sees the rank-1 ($\mathrm{SU}(2)$) asymptotic analysis of Paper 38 through Peter–Weyl rather than through the global geometry of the maximal-torus quotient G/T. (On the circle comparison: per Paper 38's shell index n and in the dual-Coxeter normalisation the $\mathrm{SU}(2)$ constant is twice the classical circle Fejér constant, while on the unit S^3 the two coincide at $2/\pi$, the factor 2 being the metric scale (Paper 38, Remark "Connection to the circle Fejér estimate"); see Section 5.2.)

Outline. Section 2 sets up the canonical compact-Lie-group spectral triple, the Peter–Weyl decomposition, the Kostant cubic Dirac operator, and the Connes–van Suijlekom truncation in Casimir-cutoff form. Section 3 proves lemmas L1′–L4 and records the fifth: §3.1 the chirality-doubled operator system substrate; §3.2 the central spectral Fejér kernel and its quantitative rate (Stein–Weiss type $\log\Lambda/\Lambda$ with constant $2/\pi$ on $\mathrm{SU}(2)$; rate form and constant open on the other simple G); §3.3 the Lipschitz comparison, proved by compression (the earlier Dirac-triangle analysis retained as representation-theoretic statements, proved on special summands and panel-verified on the rest); §3.4 the Berezin reconstruction; and §3.5 the state-space GH assembly via the approximation pair. Section 4 assembles the main theorem and states explicit corollaries for $\mathrm{SU}(N)$, $\mathrm{Spin}(n)$, $\mathrm{Sp}(n)$ and the exceptional groups. Section 5 discusses the open question whether the Fejér-moment constant is group-independent. Section 6 discusses open problems: the log-power test at higher cutoff, the extension to compact symmetric spaces G/K, the connection to Rieffel's coadjoint-orbit programme, and the Lorentzian / Wick-rotated analog.

2 Setup

2.1 Compact Lie group with bi-invariant metric

Throughout, G denotes a compact connected *semisimple* Lie group of rank $r \geq 1$, with Lie algebra $\mathfrak{g}$ and chosen maximal torus $T \subset G$ of dimension r. (Semisimplicity is where the dual-Coxeter normalisation below is defined; a central $U(1)$ factor would carry the classical circle Fejér constant rather than the $\mathrm{SU}(2)$ one, and for several simple factors the constant is open; see Remark 3.9.) We equip G with a bi-invariant Riemannian metric arising from an Ad-invariant inner product $\langle \cdot, \cdot \rangle$ on $\mathfrak{g}$. For a simple Lie algebra, this inner product is unique up to scale; we fix the normalisation in which the Casimir element acts as the dual Coxeter number on the adjoint representation,

$$C(\mathrm{ad}) \;=\; h^\vee(G), \tag{2}$$

We call this the *dual-Coxeter normalisation* throughout. *Attribution note (2026-09-03):* it is *not* the normalisation standard in representation theory and conformal field theory. That one fixes $(\theta \mid \theta) = 2$ on the highest root, under which $C(\mathrm{ad}) = (\theta \mid \theta + 2\rho) = 2 + 2(h^\vee - 1) = 2h^\vee$ [15]; the rule (2) is that form scaled by 2. The choice is immaterial to the moment constant in the cutoff variable Λ: rescaling the metric scales distances and $1/\Lambda$ together, so $\gamma_\Lambda \Lambda$ is invariant and $c(\mathrm{SU}(2)) = 2/\pi$ in every normalisation. It is material to Paper 38's per-n value: on SU(2), (2) gives the round sphere of radius 2 and $4/\pi$ per n, while $(\theta \mid \theta) = 2$ gives radius $\sqrt{2}$ and $2\sqrt{2}/\pi$, and the unit sphere gives $2/\pi$. For $G = \mathrm{SU}(N)$, $h^\vee = N$; for $\mathrm{Spin}(2n+1)$, $h^\vee = 2n-1$; for $\mathrm{Sp}(n)$, $h^\vee = n+1$; for G_2, $h^\vee = 4$, etc. For semisimple or non-simple G, we extend the normalisation factor-wise to each simple summand.

Let $\widehat{G}$ denote the set of isomorphism classes of irreducible unitary representations of G. By the Peter–Weyl theorem,

$$L^2(G) \;=\; \bigoplus_{\pi \in \widehat{G}} V_\pi \otimes V_\pi^*, \tag{3}$$

with each isotypic component $V_\pi \otimes V_\pi^*$ carrying the biregular representation of $G \times G$. For $\pi \in \widehat{G}$, we write $\dim V_\pi$ for the dimension of the representation, χ_π for its character, $C(\pi)$ for the Casimir eigenvalue in the dual-Coxeter normalisation, and λ_π for its highest weight in the standard fundamental-weight basis.

The Weyl group $W = W(G)$ acts on the Cartan subalgebra $\mathfrak{t} \subset \mathfrak{g}$ by orthogonal transformations preserving the root system. Let $\rho = \frac{1}{2}\sum_{\alpha>0} \alpha$ denote the half-sum of positive roots. The Weyl integration formula reads

$$\int_G f(g)\, dg \;=\; \frac{1}{|W|} \int_T f(t)\, |\Delta(t)|^2\, dt, \qquad \Delta(t) \;=\; \prod_{\alpha>0} \bigl(2 \sin \tfrac{\langle \alpha, t \rangle}{2}\bigr), \tag{4}$$

with dg the normalised bi-invariant Haar measure on G and dt the flat Lebesgue measure on T, and is the principal tool by which class-function computations on G reduce to Cartan integrals.

2.2 Kostant cubic Dirac operator

For G compact connected with bi-invariant metric, the canonical Dirac operator is the *Kostant cubic Dirac operator* [16], constructed from an invariant inner product on $\mathfrak{g}$ — here the dual-Coxeter-normalised bi-invariant metric, the same one in which ∇f is measured (Kostant's construction works for any invariant inner product) — via the cubic Clifford element

$$D_G \;=\; \sum_a e_a \otimes e_a \;+\; \tfrac{1}{3} \sum_{a,b,c} f_{abc}\, e_a e_b e_c, \tag{5}$$

where $\{e_a\}$ is an orthonormal basis of $\mathfrak{g}$ for that inner product and f_{abc} are the structure constants. The cubic term ensures that D_G is G-equivariant under the left regular action and self-adjoint on $L^2(G, \Sigma)$, where Σ is the spinor bundle. The spectrum of D_G^2 is given by

$$|D_G(\pi)|^2 \;=\; C(\pi) + \langle \rho, \rho \rangle \;=\; \langle \lambda_\pi + \rho, \lambda_\pi + \rho \rangle, \tag{6}$$

on each isotypic component of weight π [1, 16]. For $G = \mathrm{SU}(2)$, this reduces to the Camporesi–Higuchi spectrum $|D|^2 = (n + 1/2)^2$ used in Paper 38[2].

For our purposes, the load-bearing input is (6) and the Peter–Weyl decomposition (3); the cubic construction is invoked again only in the proof of Lemma 3.11, which uses that the cubic term is a constant (zeroth-order) Clifford element, and otherwise the proof of Theorem 1.1 is independent of the specific cubic form of D_G.

[2] The Camporesi–Higuchi convention $|\lambda_n| = n + 1/2$ on $S^3 = \mathrm{SU}(2)$ corresponds to highest weight $\lambda_\pi = n - 1$ in the standard SU(2) root system normalised by $\langle \alpha, \alpha \rangle = 2$, whence $|\lambda_\pi + \rho|^2 = (n - 1 + 1)^2/2 \cdot 2 = n^2 + n + 1/4$ up to the conventional offset.

2.3 The canonical compact-Lie-group spectral triple

We define the canonical metric spectral triple

$$\mathcal{T}_G \;=\; \big(C^\infty(G),\; L^2(G,\Sigma),\; D_G\big). \tag{7}$$

This is a unital, real spectral triple of metric dimension $\dim G$ in the sense of Connes [6] (even when $\dim G$ is even), with KO-dimension $\dim G \pmod 8$. The chirality grading γ_5 on the spinor bundle (when $\dim G$ is even) and the unitary real structure J given by charge conjugation enter as expected.

The relevant Lipschitz seminorm on $C^\infty(G)$ is the standard spectral seminorm $\|f\|_{\mathrm{Lip}} := \|[D_G, M_f]\|_{\mathrm{op}}$, where M_f denotes the multiplication operator. By the standard Kantorovich–Rubinstein-type duality arguments [6, 19], this coincides with the classical Lipschitz seminorm of f with respect to the bi-invariant geodesic distance on G; the truncated comparison constant is exactly 1 (Lemma 3.11 below).

2.4 Spectral truncation in Casimir-cutoff form

The Connes–van Suijlekom truncation [7] of $\mathcal{T}_G$ at Casimir cutoff $\Lambda > 0$ is the metric spectral triple defined as follows. Let

$$P_\Lambda \;:=\; \sum_{\pi\in\widehat{G},\, C(\pi)\le\Lambda^2} E_\pi, \tag{8}$$

where E_π is the orthogonal projection onto the isotypic component of π in $L^2(G,\Sigma)$. The truncated Hilbert space is $\mathcal{H}_\Lambda := P_\Lambda L^2(G,\Sigma)$, with dimension

$$\dim \mathcal{H}_\Lambda \;=\; \dim\Sigma \cdot \sum_{C(\pi)\le\Lambda^2} (\dim V_\pi)^2. \tag{9}$$

The truncated operator system is

$$\mathcal{O}_\Lambda \;:=\; P_\Lambda\, C^\infty(G)\, P_\Lambda \;\subset\; B(\mathcal{H}_\Lambda), \tag{10}$$

viewed as a self-adjoint operator subspace of $B(\mathcal{H}_\Lambda)$. As emphasised in [7, §1], $\mathcal{O}_\Lambda$ is *not* a C^*-subalgebra of $B(\mathcal{H}_\Lambda)$: the C^*-algebra it generates is the full bounded-operator algebra (the "C^*-envelope"), but the operator system $\mathcal{O}_\Lambda$ itself encodes finer Lipschitz-seminorm information. The truncated Dirac operator is $D_\Lambda := P_\Lambda D_G P_\Lambda$, which is block-diagonal in the Peter–Weyl decomposition and acts on each retained isotypic component by the scalar $\pm\sqrt{C(\pi)+\langle\rho,\rho\rangle}$.

The truncated metric spectral triple is then

$$\mathcal{T}_\Lambda \;=\; \big(\mathcal{O}_\Lambda,\; \mathcal{H}_\Lambda,\; D_\Lambda\big). \tag{11}$$

Recovery of Paper 38. For $G = \mathrm{SU}(2)$ and the Camporesi–Higuchi convention, the Casimir-cutoff truncation (8) agrees with the Fock-shell truncation of [26] under the bijection $n_{\max} = 2j_{\max}+1$, modulo the conventional half-integer offset. The dimension formulas and the structure of $\mathcal{O}_\Lambda$ match Paper 38's $\mathcal{O}_{n_{\max}}$. We will therefore state our results below at the level of generality of arbitrary G, with Paper 38 recovered as the rank-1 case (Corollary 4.7).

2.5 Distance convention

Throughout this paper, $d_{\mathrm{GH}}(\mathcal{T}_1,\mathcal{T}_2)$ denotes van Suijlekom's *state-space Gromov–Hausdorff distance* between metric spectral triples [8] — the object every bound in this paper controls (2026-10-02: earlier versions wrote it Λ_{prop}, a symbol inherited from the drafting lineage of Paper 38; 2026-09-02 correction: an earlier version of this subsection defined the symbol as Latrémolière's

propinquity). The strictly stronger Gromov–Hausdorff propinquity for metric spectral triples of Latrémolière [18] — the spectral-triple-level extension of the dual propinquity [17] and of the quantum compact metric space propinquity [19], computed as an infimum over tunnels — is *not* claimed: the approximation pair of §3.5 would be the data of such a tunnel, but its dual-reach hypothesis is a named gap (Remark 3.5).

3 The five lemmas

3.1 Lemma L1′: chirality-doubled operator system

The first lemma is a structural input on the operator-system substrate. Because G is parallelisable as a Lie group (the spinor bundle Σ is trivialised by left translation), the chirality grading on $L^2(G,\Sigma)$ acts diagonally on each isotypic component $V_\pi \otimes V_\pi^*$ of the Peter–Weyl decomposition. Consequently the multiplier matrix $M_f := P_\Lambda f P_\Lambda$ of any scalar $f \in C^\infty(G)$ is block-diagonal in chirality:

$$M_f \;=\; M_f^+ \oplus M_f^- \quad \text{on} \quad \mathcal{H}_\Lambda^+ \oplus \mathcal{H}_\Lambda^-, \tag{12}$$

with the two chirality blocks carrying isomorphic Weyl-sector copies of the same scalar multiplier.

Lemma 3.1 (L1′: chirality-doubled operator system substrate)**.** *The operator system $\mathcal{O}_\Lambda$ inherits from (12) a chirality block-diagonal structure. The truncated Connes distance*

$$d_{\mathcal{T}_\Lambda}(\varphi,\psi) \;=\; \sup\big\{\, |\varphi(M) - \psi(M)| \;:\; M \in \mathcal{O}_\Lambda,\ \|[D_\Lambda, M]\|_{\mathrm{op}} \le 1 \,\big\} \tag{13}$$

is finite on every cross-shell pair of pure node-evaluation states when (12) is replaced by a chirality-mixing off-diagonal extension D_Λ^{off} obtained by adding the standard $E1$-ladder couplings between adjacent Casimir shells.

Proof. The block-diagonal structure of M_f in (12) follows directly from the chirality-blindness of scalar multipliers on the parallelisable G. For the finiteness of the Connes distance on cross-shell pairs, a state pair $(\varphi_{\pi_1}, \psi_{\pi_2})$ with $\pi_1 \neq \pi_2$ has finite truncated Connes distance under the off-diagonal extension because the $E1$ ladder supplies a non-trivial structural multiplier that bounds the supremum in (13). This mechanism is verified at low rank in [26, §3.1] for SU(2) and the argument extends mechanically to any compact G via the G-equivariant ladder operators between adjacent Casimir levels (which exist in any Peter–Weyl decomposition; see for example [5, §4]).

The role of the off-diagonal extension is purely SDP-bounding: it ensures that the operator-norm constraint $\|[D_\Lambda, M]\|_{\mathrm{op}} \le 1$ is non-degenerate on the full operator system. The truthful D_Λ is what underlies the spectral-action structure of $\mathcal{T}_G$ and is what we use in Lemmas 3.11–3.26. The two operators are related by a bounded compact perturbation, and the state-space GH bound of Theorem 1.1 is robust under this choice. □

Remark 3.2. For $G = \mathrm{SU}(2)$, the off-diagonal extension is the chirality-mixing $E1$ extension of the Camporesi–Higuchi Dirac used in [26, Lemma 3.1]. For general G, the $E1$ analog is the sum of the G-equivariant raising/lowering operators between adjacent Peter–Weyl levels in the natural representation (which always exists because the natural rep contains the trivial rep among its tensor powers with itself and any other rep). The construction is universal across compact Lie groups; we do not invoke it further in the rate-controlled state-space GH proof, which uses the truthful D_Λ throughout.

3.2 Lemma L2: central spectral Fejér kernel and its moment constant

We construct the central spectral Fejér kernel $K_\Lambda \in C(G)$ generalising Paper 38's SU(2) kernel [26, Def. 2.1] to arbitrary compact G, and compute its mass-concentration moment: $(2/\pi)\log\Lambda/\Lambda$ on SU(2), with the leading constant open on the other simple groups.

Status of the group-independence question. At rank 1 the leading constant is derived: $\gamma_\Lambda(\mathrm{SU}(2)) = (2/\pi)\log\Lambda/\Lambda + O(1/\Lambda)$ **[INTERNAL THEOREM]**, which is Paper 38's $(4/\pi)\log n/n$ at $n = 2j_{\max}+1 \simeq 2\Lambda$ ($C(j) = j(j+1)$ in the dual-Coxeter rule); since $\gamma_\Lambda\Lambda$ is invariant under rescaling the metric, this value does not depend on the normalisation. Whether the leading constant is the same for every compact connected simple G is **[OPEN]**. The *Plancherel-weight × Vandermonde-Jacobian cancellation* (Remark 3.6 below) proposes a mechanism as a proof sketch; a fully rank-uniform analytical proof at the rigor level of Paper 38 Appendix A is a named gap. The numerical rank-2 and rank-3 extractions (Remark 3.7) divide by $\sqrt{C_{\mathrm{gen}}(\mathrm{ad})/h^\vee}$ while leaving the distances unscaled — on SU(2) the same pipeline returns $(2/\pi)/\sqrt{2} \approx 0.450$, not $2/\pi$ — so their agreement with $4/\pi$ compares mismatched conventions, and they discriminate neither for nor against Reading A, at G_2 included (backing test `tests/test_paper40_universal_rate.py`).

Definition 3.3 (Central spectral Fejér kernel)**.** For $\Lambda > 0$, define

$$K_\Lambda(g) \;:=\; \frac{1}{Z_\Lambda}\left|\sum_{\pi:\,C(\pi)\le\Lambda^2} \sqrt{\dim V_\pi}\,\chi_\pi(g)\right|^2, \qquad Z_\Lambda \;:=\; \sum_{\pi:\,C(\pi)\le\Lambda^2} \dim V_\pi. \tag{14}$$

The kernel is positive on G (manifestly, as a squared modulus), *central* (depends only on the conjugacy class of g), and unit-mass under the normalised bi-invariant Haar measure dg on G.

The Plancherel weight $\sqrt{\dim V_\pi}$ in (14) generalises the SU(2) factor $\sqrt{2j+1}$ used in Paper 38. The choice equidistributes the kernel amplitude's L^2 (Plancherel) mass across the retained window in proportion to $\dim V_\pi$, matching Paper 38's SU(2) kernel; see Lemma 3.4(b)–(c) below and Remark 3.5. Any normalised non-negative central kernel yields a unital completely positive smoothing map with cb-norm 1; the Plancherel weight is chosen for mass-equidistribution, not for any cb-norm property. The weight does not affect the two properties the convergence proof consumes: positivity with unit mass (a), and the mass-concentration moment $\gamma_\Lambda(G)$ (d).

Lemma 3.4 (L2: kernel construction and moment asymptotics)**.** *The central spectral Fejér kernel K_Λ of Definition 3.3 satisfies:*

(a) $K_\Lambda(g) \ge 0$ for all $g \in G$, and $\int_G K_\Lambda(g)\,dg = 1$.

(b) (Plancherel mass distribution.) The fraction of the L^2 (Plancherel) mass of the kernel amplitude $h_\Lambda := Z_\Lambda^{-1/2}\sum_{C(\pi)\le\Lambda^2}\sqrt{\dim V_\pi}\,\chi_\pi$ carried by the shell π is $\widehat{K}_\Lambda(\pi) = \dim V_\pi/Z_\Lambda$ for $C(\pi) \le \Lambda^2$, and 0 otherwise. This is not *the convolution-multiplier symbol of K_Λ: the multiplier symbol $m_{K_\Lambda}(\pi) = (\dim V_\pi)^{-1}\int_G K_\Lambda\,\overline{\chi_\pi}\,dg$ equals 1 at the trivial representation (forced by (a)), is decreasing, and is supported on the doubled window reached by Clebsch–Gordan products of retained representations.*

(c) (Smoothing map is UCP.) The smoothing map $S_{K_\Lambda} : f \mapsto K_\Lambda * f$, and its compressed Berezin form (Definition 3.25), are unital completely positive; hence $\|S_{K_\Lambda}\|_{\mathrm{cb}} = 1$ for every Λ. The quantity $\sup_\pi \widehat{K}_\Lambda(\pi) \le 2/(\Lambda+1) \to 0$ (dual-Coxeter normalisation) is the maximum of the Plancherel mass distribution of part (b); it is not the cb-norm of any map used in the proofs.

(d) (Mass-concentration moment.) Define

$$\gamma_\Lambda(G) \;:=\; \int_G K_\Lambda(g)\,d_G(e,g)\,dg, \tag{15}$$

with d_G the bi-invariant geodesic distance on G. Then $\gamma_\Lambda(G) \to 0$ (sketch in Remark 3.9), and for $G = \mathrm{SU}(2)$

$$\gamma_\Lambda(\mathrm{SU}(2)) \;=\; \frac{2}{\pi}\cdot\frac{\log\Lambda}{\Lambda} \;+\; O\big(1/\Lambda\big) \tag{16}$$

[INTERNAL THEOREM] *(Paper 38's $(4/\pi)\log n/n$ at $n \simeq 2\Lambda$; independent of the metric normalisation). For compact connected simple G of rank ≥ 2, whether $\gamma_\Lambda(G)$ has the same leading term, independent of the rank, $|W(G)|$ and the Lie type, is* ***[OPEN]*** *(Remark 3.6 proposes a mechanism as a proof sketch). For semisimple G with more than one simple factor the leading constant is open (Remark 3.9).*

Proof sketch. **(a)** The kernel is the squared modulus of a band-limited function on G weighted by $\sqrt{\dim V_\pi}$, hence non-negative. Unit mass follows from $\int_G |\chi_\pi|^2 dg = 1$ and the orthogonality of distinct χ_π in the Peter–Weyl decomposition: explicitly,

$$\int_G K_\Lambda \, dg \;=\; \frac{1}{Z_\Lambda} \sum_{\pi:\, C(\pi)\leq\Lambda^2} \dim V_\pi \cdot \int_G |\chi_\pi|^2 dg \;=\; \frac{1}{Z_\Lambda} \cdot Z_\Lambda \;=\; 1.$$

(b) The mass-distribution identity is Parseval for the amplitude h_Λ: the shell-π share of $\|h_\Lambda\|_{L^2}^2 = 1$ is $\dim V_\pi / Z_\Lambda$ by character orthogonality. The multiplier symbol is a different object: for a class function h, convolution $f \mapsto h * f$ acts on the isotypic component of π by the scalar $m_h(\pi) = (\dim V_\pi)^{-1} \int_G h\, \overline{\chi_\pi}\, dg$. Expanding the squared modulus in (14) through the Clebsch–Gordan series produces cross-terms, so $m_{K_\Lambda} \neq \widehat{K}_\Lambda$: at the trivial representation $m_{K_\Lambda}(\mathbf{1}) = \int_G K_\Lambda \, dg = 1$ by part (a), and on SU(2) at $n_{\max} = 2$ the symbol is $(1, \sqrt{2}/3, 2/9)$ — decreasing, supported on $N \leq 2n_{\max} - 1$.

(c) $K_\Lambda \geq 0$ with $\int_G K_\Lambda \, dg = 1$ makes S_{K_Λ} a unital positive convolution map; complete positivity follows because convolution against a non-negative central probability density is a point-norm limit of convex combinations of (completely isometric) translations, and a unital completely positive map satisfies $\|\varphi\|_{\mathrm{cb}} = \|\varphi(1)\| = 1$. No multiplier theorem is needed for this elementary fact; the Bożejko–Fendler multiplier theory [3] cited in earlier statements concerns uniformly bounded representations of *discrete* groups and does not apply here. The mass-distribution maximum $\sup_\pi \dim V_\pi / Z_\Lambda \leq 2/(\Lambda+1)$ follows from the Weyl dimension formula and the Casimir growth $C(\pi) \sim |\lambda_\pi + \rho|^2$, consistent with the SU(2) value $2/(n_{\max}+1)$; it plays no role in the L5 bounds.

(d) *(Moment constant.)* The mass-concentration moment $\gamma_\Lambda(G)$ admits a closed-form sum-rule via the Weyl integration formula (4). Substituting (14) into (15), expanding the squared modulus, and using the orthogonality of characters under the Weyl integration formula reduces $\gamma_\Lambda(G)$ to a finite sum over pairs of irreducibles indexed by their Casimir level. The resulting sum-rule has the Stein–Weiss / Abel–Plana structure

$$\gamma_\Lambda(G) \;=\; D_G \;-\; \frac{4\, T_\Lambda(G)}{\pi^{\dim G}\, Z_\Lambda}, \tag{17}$$

for a group-dependent diameter constant D_G and a Stein–Weiss boundary term $T_\Lambda(G)$ encoding the off-diagonal interference of the natural-coefficient Plancherel weights. At rank 1 the leading-order Stein–Weiss asymptotic of $T_\Lambda(G)/Z_\Lambda$ is computed by Abel–Plana (Paper 38, Appendix A), yielding $\gamma_\Lambda(\mathrm{SU}(2)) = (2/\pi)\cdot(\log\Lambda)/\Lambda + O(1/\Lambda)$. At rank ≥ 2 the Plancherel-weight $\times$ Vandermonde-Jacobian cancellation (Remark 3.6 below) is the proposed mechanism for the same leading term, and the leading constant is **[OPEN]**. □

Remark 3.5 (Correction note (2026-06-09))**.** The Plancherel mass distribution $\dim V_\pi / Z_\Lambda$ is distinct from the convolution-multiplier symbol of K_Λ; the smoothing map is unital completely positive with cb-norm exactly 1, and the mass-distribution maximum $2/(\Lambda+1)$ enters none of the L5 bounds. The quantities the convergence proof actually consumes — positivity and unit mass (a), the mass-concentration moment $\gamma_\Lambda(G)$ (d), its closed-form sum-rule, and its leading asymptotics (Remark 3.6) — are kernel moments, independent of the symbol distinction, and are *unaffected*. Two consequences are flagged in place: (i) the dual-reach estimate reach_P in Lemma 3.28 is reduced to a named gap — a forward cb-norm controls no partial inverse; a

corrected proof requires an antiderivative-transference lemma in the style of Leimbach–van Suijlekom [21], while for the scalar sector the conclusion is corroborated at the state-space Gromov–Hausdorff level for all compact metric groups by Gaudillot-Estrada and van Suijlekom [10]; (ii) the approximation-pair assembly of §3.5 follows Paper 38 [26]; the device is van Suijlekom's state-space Gromov–Hausdorff framework, and the tunnel vocabulary there is orientation only (2026-09-02 reframing).

Update (2026-06-10). For $G = \mathrm{SU}(2)$ the gap is closed: Paper 38 [26] now proves its theorem unconditionally by metrizing the truncations with the left-translation Lipschitz seminorm and supplying the dual direction through an exact-fit spinor lifted state (the Camporesi–Higuchi shells coincide with the $V_j \otimes V_{j\pm 1/2}$ window blocks), with both almost-inverse defects reducing to Fejér smoothings at the moment γ_Λ — no inverse estimate is used. The same translation-seminorm route is the natural repair path for general G (the lifted-state construction requires the spin-module window decomposition of the relevant Dirac shell structure per group); the general-G statement remains a named gap until that decomposition is checked case-by-case or uniformly.

Remark 3.6 (Plancherel × Vandermonde cancellation: proposed mechanism for a group-independent constant)**.** Let G be a compact connected simple Lie group with bi-invariant metric in the dual-Coxeter normalisation. The proposal is that the mass-concentration moment of the central spectral Fejér kernel of Definition 3.3 satisfies

$$\gamma_\Lambda(G) \;=\; c(G)\cdot\frac{\log\Lambda}{\Lambda} \;+\; O\big(1/\Lambda\big), \qquad c(G) = c(\mathrm{SU}(2)), \tag{18}$$

with $c(\mathrm{SU}(2)) = 2/\pi$. (Tier: at rank 1 this is Lemma 3.4(d), **[INTERNAL THEOREM]**, Paper 38; at rank ≥ 2 it is **[OPEN]**: the proof sketch below supplies a mechanism, the fully rank-uniform write-up is a named gap, and the numerical extractions do not test it — status note in §3.2. Step 2 maps Λ to a rank-1 index along each positive-root direction; whether that map is the SU(2) one, $n \simeq 2\Lambda$, for every root length is not worked out, so the value the sketch predicts at rank ≥ 2 in the variable Λ is unresolved.)

Proof sketch (mechanism; a fully rank-uniform write-up is a named gap). The argument proceeds in three steps; at rank 1 it reduces to the Paper 38 Appendix A computation, and at rank ≥ 2 it is given here at sketch level (the detailed Abel–Plana bookkeeping is not reproduced).

Step 1 (closed-form sum-rule). Substituting Definition 3.3 into (15) and expanding the squared modulus reduces $\gamma_\Lambda(G)$ to a finite double sum over pairs of irreducibles (π, π') with $C(\pi), C(\pi') \leq \Lambda^2$. The Weyl integration formula then converts the integral over G to an integral over the Cartan torus $T \subset G$ weighted by the conjugacy-class measure $|\Delta(t)|^2 dt$, where $\Delta(t) = \prod_{\alpha>0}(e^{\langle\alpha,t\rangle/2} - e^{-\langle\alpha,t\rangle/2})$ is the Weyl denominator.

Step 2 (rank-1 reduction along positive-root axes). The Weyl character formula expresses each $\chi_\pi(t)$ on the Cartan torus as a signed sum over Weyl group elements of exponential functions of $\langle w(\lambda_\pi + \rho), t\rangle$. Decomposing the Cartan-torus integration along the positive-root axes $\{t \mid \langle\alpha, t\rangle > 0\}$, the dominant contribution at each rank reduces by Pfaff–Watson partial integration to a sum of rank-1 first-moment integrals of the SU(2) type of Paper 38 — the $\sqrt{2j+1}$-weighted kernel against the $\sin^2(\chi/2)$ conjugacy-class measure — one per positive-root direction. (These are *not* the classical circle Fejér integrals $\int_{\mathbb{T}^1} F_n|\theta|$; per Paper 38's shell index n and in the dual-Coxeter normalisation the SU(2) constant is twice the classical circle Fejér constant, while on the unit S^3 the two coincide at $2/\pi$, the factor 2 being the metric scale (Paper 38, Remark "Connection to the circle Fejér estimate").)

Step 3 (Plancherel × Vandermonde cancellation). The central spectral Fejér kernel is built with Plancherel weights $\sqrt{\dim V_\pi}$ (Definition 3.3). When γ_Λ is computed via the Weyl integration formula, the Plancherel weights appear squared as $\dim V_\pi$, and the Vandermonde Jacobian $|\Delta(t)|^2$ appears as a multiplicative factor on the Cartan torus. The Weyl dimension

formula expresses

$$\dim V_\pi \;=\; \prod_{\alpha>0} \frac{\langle \lambda_\pi + \rho,\, \alpha\rangle}{\langle \rho,\, \alpha\rangle},$$

which has the same Vandermonde-style $\prod_{\alpha>0}$ structure as $|\Delta(t)|^2$. The proposal (unproved) is that these two products cancel at leading order, which would eliminate all rank-, $|W|$-, and N_+-dependence (where N_+ is the number of positive roots). What would remain is the rank-1 (SU(2)) constant of Paper 38, Appendix A ($4/\pi$ in its index n, $2/\pi$ in Λ at rank 1), from each positive-root direction's Euler–Catalan asymptote $\sum_{d\ \mathrm{odd}} 1/d^2 = \pi^2/8$; the value this predicts in Λ at rank ≥ 2 is unresolved (statement above).

The detailed Abel–Plana bookkeeping required to convert the leading-order asymptotic into a uniform $O(1/\Lambda)$ remainder is standard but tedious at general rank; carrying it out to the rigor level of the rank-1 Stein–Weiss analysis in Paper 38 Appendix A, uniformly in rank, is a named gap. At rank ≥ 2 the numerical extractions below do not corroborate the cancellation (status note in §3.2). □

Numerical extraction. The rank-2 and rank-3 extractions are recorded below; they do not test the proposed mechanism.

Remark 3.7 (Rank-≥ 2 extractions of the leading constant; group-independence **[OPEN]**). The extraction pipeline fits $\Lambda\gamma_\Lambda(G) = c\log\Lambda + b$ on the asymptotic tail of each panel and divides the result by $\sqrt{C_{\mathrm{gen}}(\mathrm{ad})/h^\vee}$, leaving the distances unscaled. Its outputs are:

Group	rank	pipeline output	panel (Λ^2)	notes
SU(2)	1	$0.450 = (2/\pi)/\sqrt{2}$	—	anchor $2/\pi$, Lemma 3.4(d)
SU(3)	2	1.243	$[1, 1000]$	
Sp(2) = Sp(4)	2	1.087	$[1, 1000]$	fit-unstable
G_2	2	1.177	$[1, 1000]$	
SU(4)	3	0.900	$[1, 300]$	compute-budget; see Remark 3.8

Since $\gamma_\Lambda\Lambda$ is invariant under rescaling the metric, the division is not a change of convention but a group-dependent error, as the SU(2) row shows. The outputs are therefore comparable neither with the SU(2) anchor nor with $4/\pi$: their proximity to $4/\pi$ at every rank, and the Reading A versus Reading B comparison built on it (including the G_2 gap and the forced-coefficient χ^2 comparison at SU(4)), compare mismatched conventions and decide nothing. Whether $c(G)$ is group-independent is **[OPEN]**; the rank-≥ 2 values in the anchor's convention are not computed here. Three-parameter Stein–Weiss fits scatter widely at Sp(2) (`tests/test_paper40_universal_rate.py`). The computation uses the generic compact-Lie-group infrastructure (Cartan matrix, Freudenthal weight enumeration, Brauer–Klimyk tensor decomposition, Weyl integration formula, 2D Gauss–Legendre quadrature on $T \subset G$ at rank 2 and 3D at rank 3), documented in the supporting computational memos [29, 31].

Remark 3.8 (Compute scaling at rank 3). The SU(4) extraction carries the convention mismatch of Remark 3.7, and two further factors degrade it: (i) the rank-3 Weyl integration is genuinely 3-dimensional, increasing per-row compute cost by roughly an order of magnitude over rank 2 and capping the practical panel at $\Lambda^2 = 300$; (ii) the pre-asymptotic structure is stronger at rank 3 than at rank 2, making the leading-order extraction more sensitive to the fit form. See [31] for the detailed analysis.

Remark 3.9 (Several simple factors: the leading constant is open). *Withdrawn 2026-10-02.* An earlier version of this paper stated, as a corollary, that $c(G) = 4/\pi$ extends from simple to semisimple G "via factor-wise cancellation", arguing that on $G_1 \times G_2$ the kernel factors, the moment is additive, and the constants combine as $c(G_1)/2 + c(G_2)/2 = 4/\pi$. Each step fails.

The Casimir cutoff $C_1(\pi_1) + C_2(\pi_2) \le \Lambda^2$ is a joint window, so the kernel is not the product of the single-factor kernels at cutoff Λ; and even for a product kernel the first moment of $d = (d_1^2 + d_2^2)^{1/2}$ is not additive — it lies strictly between $\max_i \int d_i$ and $\sum_i \int d_i$. Paper 39 measures the second fact on the same-manifold product $\mathrm{SU}(2) \times \mathrm{SU}(2)$: at cutoffs $(2, 2)$ its joint moment is about 0.74 of the sum of the two factor moments, about 1.5 times one factor's (`tests/test_paper39_lifted_state.py`). No value of the leading constant is asserted for semisimple G with more than one simple factor. What survives for such G is the bound $d_{\mathrm{GH}} \le \gamma_\Lambda(G)$ of Theorem 4.1, at that theorem's tier, with $\gamma_\Lambda(G) \to 0$ (sketch: the Casimir window is a ball in the weight lattice whose boundary shells carry a vanishing share of Z_Λ, so the multiplier symbol of K_Λ tends to 1 on each fixed irreducible, the kernel tends weak-$*$ to δ_e, and its moment of the continuous function $d_G(e, \cdot)$ tends to 0); the rate is not established here.

Scope (2026-09-02). An earlier version of the withdrawn corollary extended the statement to $G_{\mathrm{ss}} \times T^k$ with a central torus factor, on the ground that "the rank-1 constant is $4/\pi$ on each circle". That ground is false: on a circle factor the central Fejér kernel reduces to the classical Fejér kernel, whose probability-normalised first moment is $(2/\pi) \log n/n$ — half Paper 38's per-n SU(2) value (Paper 38, Remark "Connection to the circle Fejér estimate"; `tests/test_trunk_qa_fejer_4_over_pi.py`). Even under the withdrawn factor-wise argument the product constant would depend on the mix of simple and circle factors rather than being group-independent, so the torus-factor extension is withdrawn; no value is asserted here for $c(G_{\mathrm{ss}} \times T^k)$.

Remark 3.10 (Relation to the circle Fejér constant)**.** Paper 38's per-n form of (16), $(4/\pi) \log n/n$ in the dual-Coxeter normalisation, has *twice* the constant of the classical estimate for the probability-normalised Fejér kernel on the circle [25]: $\frac{1}{2\pi} \int_{\mathbb{T}^1} F_n(\theta)\, |\theta|\, d\theta \sim (2/\pi) \log n/n$ (Paper 38, Remark "Connection to the circle Fejér estimate", where the circle constant is computed in closed form and measured against the SU(2) value: `tests/test_trunk_qa_fejer_4_over_pi.py`). The $\sqrt{\dim V_\pi}$ Plancherel weighting in Definition 3.3 produces the central Fejér-type (squared-Dirichlet; on the circle, the Cesàro-1 Fejér) spectral kernel, the compact-group analogue of the spectral Fejér kernel of [21], which states no rate; the logarithm is carried by the circle's Fejér kernel (Paper 38 Remark rem:circle_fejer). No common "2Vol(base)/Vol(group)" form is claimed (Paper 38): on the unit-radius spheres the SU(2) quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)) = 2/\pi$ equals the circle constant, and $4/\pi$ is its dual-Coxeter per-n value; whether each simple group sees the rank-1 (SU(2)) constant through the Plancherel × Vandermonde cancellation is **[OPEN]**, discussed further in §5.

3.3 Lemma L3: Lipschitz comparison by compression

The third lemma is the Lipschitz comparison estimate that controls the operator norm of the commutator $[D_G, M_f]$ in terms of the classical Lipschitz norm of f on G. It is proved at every rank by compression, exactly as Paper 38 now proves the SU(2) case (2026-10-02; an earlier version of this section proved it through a higher-rank analog of Paper 38's Avery 3-Y selection rule). The *Dirac-triangle inequality* on tensor products of irreducibles, which earlier versions used as the proof, is retained below (§3.3.2–§3.3.3) as a statement about Kostant eigenvalues and Casimirs of tensor-product summands; its bridge to the commutator norm (an entrywise bound read as an operator-norm bound, and eigenvalue ratios read as per-isotypic Lipschitz constants) is the step Paper 38 refuted at rank 1 on 2026-09-28 and is withdrawn here.

3.3.1 Statement and proof by compression

Lemma 3.11 (L3: Lipschitz comparison on compact Lie groups; **[INTERNAL THEOREM]**)**.** *Let G be a compact connected Lie group of rank $r \ge 1$ with bi-invariant metric (dual-Coxeter normalised), and let D_G be the Kostant cubic Dirac operator on G. For every real*

$f \in C^\infty(G)$ and every $\Lambda > 0$,

$$\|[D_\Lambda,\, P_\Lambda M_f P_\Lambda]\|_{\mathrm{op}} \;\le\; C_3(G)\cdot \|\nabla f\|_{L^\infty(G)}, \qquad C_3(G) = 1, \tag{19}$$

where the Lipschitz norm $\|\nabla f\|_{L^\infty}$ is taken with respect to the bi-invariant Riemannian metric on G. The constant cannot be lowered uniformly in Λ.

Proof (by compression). (a) The truncation commutes with D_G. The projection E_π of (8) is the isotypic projection of the left regular action of G on $L^2(G,\Sigma) = L^2(G)\otimes\Sigma$ (the spinor bundle trivialised by a left-invariant frame), namely $E_\pi = \dim\pi \int_G \overline{\chi_\pi(g)}\, L_g\, dg$ (in $L^2(G) = \bigoplus_\pi V_\pi \otimes V_\pi^*$ the left- and right-isotypic components coincide, with equal Casimirs). D_G is equivariant under the left regular action (§2.2), so it commutes with every L_g, hence with every E_π and with P_Λ. Therefore $D_\Lambda = P_\Lambda D_G P_\Lambda = D_G P_\Lambda$ and

$$[D_\Lambda,\, P_\Lambda M_f P_\Lambda] \;=\; P_\Lambda [D_G, M_f] P_\Lambda.$$

(b) The continuum commutator. In (5) the cubic term is a constant Clifford element acting on the spinor factor alone, so it commutes with M_f; the first-order term $\sum_a e_a \otimes X_a$ (the X_a the left-invariant orthonormal frame fields) gives $[D_G, M_f] = \sum_a e_a\,(X_a f) = c(df)$, Clifford multiplication by df. For real f its operator norm is $\sup_G |df| = \|\nabla f\|_{L^\infty(G)}$, the Clifford relations being those of the bi-invariant metric. (This is Connes' formula on a closed spin manifold; the cubic torsion term does not enter.)

(c) Compression contracts norms, so $\|P_\Lambda[D_G, M_f]P_\Lambda\|_{\mathrm{op}} \le \|\nabla f\|_{L^\infty(G)}$ for every f and every Λ. *Sharpness:* $P_\Lambda \to 1$ strongly as $\Lambda\to\infty$, so $\|P_\Lambda[D_G, M_f]P_\Lambda\|_{\mathrm{op}} \to \|[D_G, M_f]\|_{\mathrm{op}}$ and $C_3(G) = 1$ cannot be lowered uniformly. □

The argument uses only left-equivariance of D_G and that its symbol is Clifford multiplication, so it holds verbatim for every compact connected Lie group in the class, at every rank, with no case analysis; it replaces the per-summand route below as the proof of Lemma 3.11. Mechanism backed by `tests/test_paper40_l3_compression.py`.

The key analytical ingredient of the per-summand route is the Dirac-triangle inequality, which generalises the SU(2) selection rule $|n - n'| + 1 \le N \le n + n' - 1$ used in Paper 38 to arbitrary rank.

Definition 3.12 (Dirac-triangle inequality)**.** Let G be a compact simply-connected simple Lie group. For an irreducible representation π of G with highest weight λ_π, write $D(\pi) := |D_G(\pi)| = \sqrt{\langle \lambda_\pi + \rho, \lambda_\pi + \rho\rangle}$ for the Kostant Dirac eigenvalue. The *Dirac-triangle inequality* (DT) states that for every triple π, π', σ with $\sigma \subset \pi\otimes\pi'^*$ in the Brauer–Klimyk tensor-product decomposition,

$$|D(\pi) - D(\pi')| \;\le\; \sqrt{C(\sigma)}. \tag{20}$$

Lemma 3.13 (Dirac-triangle reduction)**.** *The Dirac-triangle inequality* (20) *follows from the* intermediate inequality *(INT)*

$$|\lambda_\pi - \lambda_{\pi'}|^2 \;\le\; C(\sigma) \quad \textit{for every } \sigma \subset \pi\otimes\pi'^*, \tag{21}$$

by the reverse triangle inequality for vector norms.

Proof. $D(\pi) = |\lambda_\pi + \rho|$ is the norm of a real vector. By the reverse triangle inequality,

$$\big|\,|\lambda_\pi + \rho| - |\lambda_{\pi'} + \rho|\,\big| \;\le\; |(\lambda_\pi + \rho) - (\lambda_{\pi'} + \rho)| \;=\; |\lambda_\pi - \lambda_{\pi'}|.$$

Squaring preserves the inequality. Combining with (INT) gives (DT). □

3.3.2 Numerical verification of (DT) and (INT)

The Dirac-triangle inequality has been verified across compact simple Lie groups of types A, C, and G on a comprehensive panel:

Proposition 3.14 (Numerical verification, 977/977 panel pass)**.** *The Dirac-triangle inequality* (20) *and the intermediate inequality* (21) *both hold without exception on the following panels:*

Panel	*Ordered pairs*	*(DT) max ratio*	*(INT) max ratio*
SU(3), $p+q \le 5$	*441*	*0.763*	*0.625*
SU(4), $a+b+c \le 3$	*400*	*0.625*	*0.429*
Sp(2) = Sp(4), $a+b \le 3$	*100*	*0.694*	*0.500*
G_2, $a+b \le 2$	*36*	*0.628*	*0.400*
Total	***977***	—	—

All ratios computed in exact sympy arithmetic in the dual-Coxeter normalisation; the worst-case panel ratio is 0.763 *for (DT) and* 0.625 *for (INT). No counterexamples found.*

Verification. Performed via the generic compact-Lie-group infrastructure documented in the supporting memo [30]: Cartan matrices and Gram matrix on the fundamental-weight basis, Freudenthal's recursion for weight multiplicities, Brauer–Klimyk (Racah–Speiser) tensor product decomposition, all in exact sympy arithmetic. The asymptotic family $\lambda_n = (n, 0, \ldots, 0)$, $\lambda' = (1, 0, \ldots, 0)$ gives the cleanest convergence-rate check: the ratio $|D(\lambda_n) - D(\lambda')|/\sqrt{C(\sigma_{\min})}$ approaches 1 from below at the rate $1 - O(1/n)$ in every rank tested. (An earlier version called this structurally identical to a per-harmonic ratio $\sqrt{(N-1)/(N+1)}$ of Paper 38; that per-harmonic inequality is false and was withdrawn in Paper 38, 2026-09-28.) □

3.3.3 The per-summand inequalities: (INT) on special summands, PRV existence, and an interior-summand sketch

Status of the per-summand analysis. Lemma 3.11 is proved above by compression, so nothing below is needed for it. What follows treats the Dirac-triangle (DT) and intermediate (INT) inequalities as statements about Kostant eigenvalues and Casimirs (proved for the special summand classes (i)–(ii) below, panel-verified for (iii)–(iv)); its bridge to the commutator norm (an entrywise bound read as an operator-norm bound, and eigenvalue ratios read as per-isotypic Lipschitz constants) is the step Paper 38 refuted at rank 1 on 2026-09-28 and is withdrawn here, so the statements below about "C_3" are to be read as statements about (DT)/(INT), not as Lipschitz constants. By summand class $\sigma \subset \pi \otimes \pi'^*$: (INT) is proved at all ranks for (i) the trivial-λ case and (ii) the Cartan-product summand (Proposition 3.15); for (iii) the PRV (Parthasarathy–Ranga Rao–Varadarajan) summands, Theorem 3.16 gives existence at all ranks, but its inequality (b) falls short of (INT) by $|\rho|^2$, so (INT) on that class rests on the exact-arithmetic panels at A_2, A_3, C_2, G_2; and (iv) the interior (non-PRV) summands are covered on those weight-bounded panels by Lemma 3.18 (the higher exceptional ranks a named follow-on, below). The PRV class contains the asymptotic family $\lambda_n = (n, 0, \ldots, 0)$ tensored with $\lambda' = (1, 0, \ldots, 0)^*$, along which the (DT) ratio tends to 1 in every rank tested (a statement about eigenvalues). The closure of interior (non-PRV) summands via the Brauer–Klimyk signed-sum cancellation mechanism is sketched analytically in Lemma 3.18 (a proof sketch at general rank), and (INT) on interior summands is verified in exact sympy arithmetic at 5641/5641 cells across SU(3), SU(4), Sp(2), and G_2 (extended panel; see Remark 3.21). The combined status (Corollary 3.22) is therefore: (INT) at all ranks for classes (i)–(ii), and on the tested panels for classes (iii)–(iv). The *interior* (INT) leg of the per-summand route is verified by exhaustive numeric weight-bounded panels at A_2, A_3, C_2, G_2, with an analytical sketch; its rank-uniform

case-bookkeeping for the higher exceptional ranks (F_4, E_6, E_7, E_8) carries through the same analytical skeleton but is a named follow-on (Remark 3.21).

Proposition 3.15 (Rigorous proofs of (INT) at low rank and special summands)**.** *The intermediate inequality (21) holds rigorously in the following cases:*

(i) ***Trivial*** λ*.* *If $\lambda_\pi = 0$, then $V_\pi \otimes V_{\pi'}^* = V_{\pi'^*}$ and the unique summand $\sigma = \pi'^*$ has $C(\sigma) = C(\pi') \geq |\lambda_{\pi'}|^2$, with the inequality $C(\pi') \geq |\lambda_{\pi'}|^2$ following from $C(\lambda) = \langle \lambda, \lambda + 2\rho \rangle = |\lambda|^2 + 2\langle \lambda, \rho \rangle \geq |\lambda|^2$ since λ is dominant and ρ is regular dominant, with all inverse-Cartan entries non-negative.*

(ii) ***Cartan-product summand.*** *The Cartan product $\sigma_{\max} = \lambda_\pi + \lambda_{\pi'^*}$ always appears with multiplicity 1 in $V_\pi \otimes V_{\pi'}^*$, and $C(\sigma_{\max}) \geq |\lambda_\pi - \lambda_{\pi'}|^2$ holds by an explicit Cauchy–Schwarz computation: $|\lambda_\pi + \lambda_{\pi'^*}|^2 - |\lambda_\pi - \lambda_{\pi'}|^2 = 2\langle \lambda_\pi, \lambda_{\pi'} + \lambda_{\pi'^*} \rangle \geq 0$ by dominance and non-negativity of inverse-Cartan entries.*

Theorem 3.16 (PRV summands: existence at all ranks, and an inequality weaker than (INT))**.** *Let G be a compact connected simple Lie group with bi-invariant metric in the dual-Coxeter normalisation, and let π, π' be irreducible representations of G. For each w in the Weyl group $W = W(G)$, let*

$$\sigma_w \;:=\; \overline{\lambda_\pi + w\,\lambda_{\pi'}^*} \tag{22}$$

denote the dominant image (under the Weyl chamber projection) of $\lambda_\pi + w\,\lambda_{\pi'}^*$. Then:

(a) (Existence, Kumar's theorem.) For every $w \in W$, the representation σ_w appears as a summand of $V_\pi \otimes V_{\pi'}^*$ in the Brauer–Klimyk decomposition.

(b) (Inequality.) $|\sigma_w + \rho|^2 \geq |\lambda_\pi - \lambda_{\pi'}|^2$ at every PRV summand σ_w. (Corrected 2026-10-02: since $C(\sigma) = |\sigma + \rho|^2 - |\rho|^2$, this yields the intermediate inequality (21) only in the stronger form $|\sigma_w + \rho|^2 \geq |\lambda_\pi - \lambda_{\pi'}|^2 + |\rho|^2$, which the sketch below does not establish; (INT) on the PRV class rests on the exact-arithmetic panels of Proposition 3.14.)

Proof sketch; detailed bookkeeping in [30]. Part (a) is the Parthasarathy–Ranga Rao–Varadarajan conjecture, proved by Kumar [32] (and independently by Mathieu [33]). Part (b) follows from the standard dominance-order fact [35, §13]: for any element v of the Cartan subalgebra and any $w \in W$, the dominant image $\overline{w \cdot v}$ in the closed fundamental Weyl chamber satisfies $\langle \overline{w \cdot v}, \rho \rangle \geq \langle w \cdot v, \rho \rangle$. Applying this with $v = \lambda_\pi + w \cdot \lambda_{\pi'}^*$ and expanding $|\sigma_w + \rho|^2 - |\lambda_\pi - \lambda_{\pi'}|^2$ gives a sum of inner products that is non-negative by the standard Weyl-dominance lemma (the dominant Weyl-orbit representative maximises $\langle \cdot, \rho \rangle$; Bourbaki [4]) and the dual-Coxeter positivity of the inverse-Cartan entries. The bound is type-independent: the proof goes through for every simple Lie algebra type $(A_n, B_n, C_n, D_n, G_2, F_4, E_6, E_7, E_8)$. Detailed bookkeeping is given in [30], §§3–6. □

Corollary 3.17 ((DT) ratio along the asymptotic family)**.** *The PRV class (Theorem 3.16(a)) contains the asymptotic family $\pi_n = (n, 0, \ldots, 0)$ tensored with $\pi' = (1, 0, \ldots, 0)^*$. Along this family the (DT) ratio tends to 1 (a statement about eigenvalues and Casimirs, checked in every rank tested; it is not a bound on the Lipschitz constant, which Lemma 3.11 gives by compression).*

Proof. The asymptotic family $\pi_n \otimes \pi'^*$ decomposes (at large n) into a small number of summands all of which are PRV summands of the form $\sigma_w = \overline{(n, 0, \ldots, 0) + w \cdot (1, 0, \ldots, 0)^*}$ for w running over a finite subset of W. On this family the intermediate inequality (21) is verified in exact arithmetic at the tested ranks (Proposition 3.14; Theorem 3.16(b) alone does not give it), and the per-isotypic ratio $|D(\pi_n) - D(\pi')|^2 / C(\sigma_w)$ approaches 1 from below as $n \to \infty$ at the rate $1 - O(1/n)$ in every rank tested (Proposition 3.14). (An earlier version read this ratio as a per-isotypic Lipschitz constant $C_3(G, \sigma)$ with supremum 1; that reading is the withdrawn bridge.) □

The remaining summands σ of $V_\pi \otimes V_{\pi'}^*$ that are not PRV summands are referred to as *interior summands*. (21) on this class is panel-verified and sketched analytically in the next lemma, which extracts the load-bearing structural fact underlying the Brauer–Klimyk signed-sum cancellation mechanism and states it as a single positivity inequality on the positive-root cone $Q^+ \subset \mathfrak{h}^*$.

Lemma 3.18 (L3-interior: (INT) on interior summands via Brauer–Klimyk signed-sum closure; **[PANEL-VERIFIED]** at A_2, A_3, C_2, G_2, a proof sketch at general rank)**.** *Let G be a compact connected simple Lie group with bi-invariant metric in the dual-Coxeter normalisation, let π, π' be irreducible representations of G with highest weights $\lambda := \lambda_\pi$ and $\lambda' := \lambda_{\pi'}$, and let $\sigma \subset V_\pi \otimes V_{\pi'}^*$ be an interior (non-PRV) summand of the Brauer–Klimyk decomposition appearing with positive multiplicity. Then σ satisfies the intermediate inequality*

$$|\lambda - \lambda'|^2 \;\le\; C(\sigma). \tag{23}$$

Verified in exact arithmetic on the panels of Remark 3.21; at general rank the argument below is a sketch, since Step 4's cancellation is asserted and its case-bookkeeping is tabulated only on those panels.

Proof. The proof reduces (23) to a single positivity statement on the positive-root cone, and argues the latter by combining the weight-containment property of every contributing summand (Lemma 1 below) with the standard Steinberg signed-sum formula for tensor-product multiplicities.

Step 1 (Steinberg signed-sum positivity). The Steinberg formula for Littlewood–Richardson coefficients on a compact simple Lie group (see [9, Ch. 24 §4] or [4, Ch. VIII §9] for the textbook statement) reads

$$N^{\sigma}_{\pi,\pi'^*} \;=\; \sum_{w \in W} \operatorname{sgn}(w)\, \dim V_{\pi'^*}\big[w \cdot (\sigma + \rho) - \lambda - \rho\big], \tag{24}$$

where $\dim V_{\pi'^*}[\nu]$ denotes the multiplicity of the weight ν in $V_{\pi'^*}$. By hypothesis $N^{\sigma}_{\pi,\pi'^*} > 0$, so the signed sum on the right-hand side is strictly positive. In particular there exists at least one $w_* \in W$ for which

$$\mu^* \;:=\; w_*(\sigma + \rho) - \lambda - \rho \tag{25}$$

is a weight of $V_{\pi'^*}$ with $\operatorname{sgn}(w_*) \dim V_{\pi'^*}[\mu^*] \neq 0$ and whose contribution is not cancelled by any companion Weyl element of opposite sign in the orbit decomposition. We fix such a w_* once and for all.

Step 2 (weight-containment via Lemma 1). For any dominant σ appearing with positive multiplicity in $V_\pi \otimes V_{\pi'^*}$, the weight $\sigma - \lambda$ lies in the support of $V_{\pi'^*}$. Indeed, the tensor-product weight-multiplicity formula

$$\dim(V_\pi \otimes V_{\pi'^*})[\sigma] \;=\; \sum_{\nu \in \operatorname{wts}(V_\pi)} \dim V_\pi[\nu]\, \dim V_{\pi'^*}[\sigma - \nu]$$

contains the term $\nu = \lambda$ (highest-weight contribution with multiplicity 1), and this term is non-zero exactly when $\sigma - \lambda$ is a weight of $V_{\pi'^*}$. Standard highest-weight-vector machinery (see e.g. [9, Prop. 25.30]) shows that this contribution survives the highest-weight-vector decomposition, so

$$\sigma - \lambda \;\in\; \operatorname{wts}(V_{\pi'^*}). \tag{26}$$

The lowest weight of $V_{\pi'^*}$ is $-\lambda'$, hence $\sigma - \lambda \ge -\lambda'$ in the standard partial order on $\mathfrak{h}^*$, equivalently

$$Q \;:=\; \sigma + \lambda' - \lambda \;\in\; Q^+, \qquad Q^+ := \mathbb{Z}_{\ge 0}\Delta^+ \tag{27}$$

(the non-negative integer cone over the positive simple roots).

Step 3 (algebraic reduction to (SI′)). Substituting $\sigma = \lambda - \lambda' + Q$ in the squared norm $|\sigma+\rho|^2$ and expanding via the inner product:

$$\begin{aligned}|\sigma+\rho|^2 - |\lambda-\lambda'+\rho|^2 &= |\lambda-\lambda'+\rho+Q|^2 - |\lambda-\lambda'+\rho|^2 \\ &= 2\langle \lambda-\lambda'+\rho,\, Q\rangle + |Q|^2.\end{aligned}$$

Using $\sigma+\rho = \lambda-\lambda'+\rho+Q$, this rearranges to

$$|\sigma+\rho|^2 - |\lambda-\lambda'+\rho|^2 \;=\; 2\langle \sigma+\rho,\, Q\rangle - |Q|^2. \tag{28}$$

The inequality (23) is equivalent, after expanding $C(\sigma) = |\sigma+\rho|^2 - |\rho|^2$, to $|\sigma+\rho|^2 \geq |\lambda - \lambda'|^2 + |\rho|^2$, and since $|\lambda-\lambda'+\rho|^2 = |\lambda-\lambda'|^2 + 2\langle\lambda-\lambda',\rho\rangle + |\rho|^2$, it suffices to prove the stronger statement

$$2\langle \sigma+\rho,\, Q\rangle \;\geq\; |Q|^2, \qquad Q = \sigma+\lambda'-\lambda \in Q^+ \tag{SI′}$$

together with the auxiliary $\langle\lambda-\lambda',\rho\rangle \geq 0$ correction. (The latter is automatic for $\lambda \geq \lambda'$ in the dominance order, and is absorbed into the case-bookkeeping otherwise; see [30, §4.3].) We now prove (SI′).

Step 4 (Brauer–Klimyk signed-sum cancellation of small-norm candidates). The Steinberg formula (24) expresses N^{σ}_{π,π'^*} as a signed sum over W. For each $w \in W$, the candidate weight $\mu_w := w(\sigma+\rho) - \lambda - \rho$ is either a weight of $V_{\pi'^*}$ (in which case it contributes $\operatorname{sgn}(w)\dim V_{\pi'^*}[\mu_w]$ to the sum) or is not (in which case it contributes 0). Group the contributing Weyl elements into pairs (w, w') with $\operatorname{sgn}(w) = -\operatorname{sgn}(w')$ and $w(\sigma+\rho), w'(\sigma+\rho)$ in the same Weyl orbit of $\lambda+\rho+\operatorname{wts}(V_{\pi'^*})$. Each such pair contributes $\operatorname{sgn}(w)(\dim V_{\pi'^*}[\mu_w] - \dim V_{\pi'^*}[\mu_{w'}])$ to the sum.

The load-bearing structural observation is the following. Define the *Brauer–Klimyk admissible set*

$$\mathcal{A}(\sigma) \;:=\; \big\{\mu^* \in \operatorname{wts}(V_{\pi'^*}) : \exists\, w \in W,\, w_*(\sigma+\rho) = \lambda+\mu^*+\rho,\, \operatorname{sgn}(w)\,\dim V_{\pi'^*}[\mu^*] \text{ is not cancelled}\big\}.$$

For every $\mu^* \in \mathcal{A}(\sigma)$, Weyl-invariance of the norm gives

$$|\sigma+\rho|^2 \;=\; |\lambda+\mu^*+\rho|^2. \tag{29}$$

We claim that the signed-sum cancellation mechanism precisely removes those candidate μ^* for which the right-hand side of (29) would yield a violation of (SI′).

To see this, fix $w \in W$ and write $\mu_w = w(\sigma+\rho) - \lambda - \rho$. Substituting in (28):

$$\begin{aligned}2\langle\sigma+\rho, Q\rangle - |Q|^2 &= |\lambda+\mu_w+\rho|^2 - |\lambda-\lambda'+\rho|^2 \\ &= 2\langle\lambda+\rho, \mu_w+\lambda'\rangle + |\mu_w|^2 - |\lambda'|^2.\end{aligned}$$

For this quantity to be non-negative, we need

$$2\langle\lambda+\rho, \mu_w+\lambda'\rangle + |\mu_w|^2 \;\geq\; |\lambda'|^2. \tag{30}$$

The candidates μ_w that *violate* (30) are those for which μ_w is in the interior of the convex hull of $W\cdot(-\lambda')$ — i.e., neither $\mu_w = -\lambda'$ nor in the Weyl orbit of $-\lambda'$. By Lemma 1, the contributing $\mu^* \in \mathcal{A}(\sigma)$ must be of the special form $\sigma - \lambda$, which by Step 2 satisfies $|\sigma-\lambda|^2 \leq |\lambda'|^2$ (norm-decrease under the weight-containment $\sigma - \lambda \in \operatorname{wts}(V_{\pi'^*})$). The Steinberg signed sum then cancels exactly those orbit elements that would violate (30).

Step 5 (closure via σ-dominance and $\lambda+\rho$ regularity). Combining Step 2, Step 3, and the cancellation argument of Step 4, the surviving contribution from $\mu^* \in \mathcal{A}(\sigma)$ satisfies (30). Substituting back via (29) and (28) gives $2\langle\sigma+\rho, Q\rangle \geq |Q|^2$, which is (SI′). Hence, granted Step 4's cancellation (tabulated on the panels, Remark 3.19), (23) holds on every interior summand. □

Remark 3.19 (Per-orbit case-bookkeeping)**.** The proof above reduces (23) to the per-witness inequality (30) together with the Brauer–Klimyk cancellation of small-norm orbit elements. The per-orbit case-by-case enumeration is finite at each rank (bounded by $|W|^2$) and follows the same Weyl-orbit decomposition used in the asymptotic-family analysis of Corollary 3.17. The detailed enumeration was carried out across $\mathrm{SU}(3), \mathrm{SU}(4), \mathrm{Sp}(2)$, and G_2 in the supporting memo [30, §§4–5], exhibiting 5641 surviving interior summands across the four panels and confirming that the per-orbit pairing predicted by the Brauer–Klimyk sign rule is non-degenerate at every cell. We treat this enumeration as the concrete content of Step 4's cancellation argument and refer the reader to the memo for the per-cell tables.

Remark 3.20 (Why the signed-sum mechanism is expected to succeed at general rank)**.** The Brauer–Klimyk signed-sum cancellation mechanism is the structural ingredient of the per-summand sketch for interior summands (Lemma 3.11 itself is proved by compression). On the PRV class (Theorem 3.16) existence needs no signed-sum cancellation, and the inequality (b) from the standard dominance-order fact [35, §13] applied to the rho-shift falls short of (INT) by $|\rho|^2$. The interior case (Lemma 3.18) is argued via the same algebraic ingredients applied through the Steinberg formula, with the additional mechanism that contributions from candidate weights μ_w violating the per-witness inequality (30) are precisely those cancelled by the alternating-sign sum. Both arguments share the common skeleton: Lemma 1 (weight-containment), the Weyl-dominance lemma (the dominant orbit representative maximises $\langle \cdot, \rho \rangle$; Bourbaki [4]), and the dual-Coxeter positivity of the inverse-Cartan matrix. The Brauer–Klimyk cancellation adds the pairing structure the sketch uses to extend the per-witness inequality to all interior summands; that extension is verified on the panels only.

Remark 3.21 (Numerical certification at the tested panels)**.** The intermediate inequality (23) on interior summands has been verified in exact sympy arithmetic at every cell of the extended panel of 5641 interior summands across $\mathrm{SU}(3)$ ($p{+}q \le 5$), $\mathrm{SU}(4)$ ($a{+}b{+}c \le 3$), $\mathrm{Sp}(2)$ ($a{+}b \le 3$), and G_2 ($a + b \le 2$), with no exception. This numerical certification (the extended panel, beyond the 977-pair panel of Proposition 3.14) provides a complete check of the per-orbit case-bookkeeping of Remark 3.19 on the four panels. This exhaustive verification establishes (23) on these weight-bounded panels (not on whole ranks), at the rigor level of Paper 38 Appendix A; the analytical sketch of Lemma 3.18 is what would extend it.[3]

Corollary 3.22 ((DT)/(INT) status by summand class)**.** *(INT) holds at all ranks for the trivial-λ and Cartan-product summands (Proposition 3.15). On the PRV class (Theorem 3.16(a)) and the interior class (Lemma 3.18) it is established on the exact-arithmetic panels at A_2, A_3, C_2, G_2; Theorem 3.16(b) falls short of (INT) by $|\rho|^2$, and the interior case-bookkeeping at F_4, E_6, E_7, E_8 is a named follow-on. Where (INT) holds, Lemma 3.13 gives the Dirac-triangle inequality (20) (a representation-theoretic statement). Neither bounds C_3: that bridge is withdrawn, and Lemma 3.11 is proved by compression.*

Withdrawn per-summand bridge from (DT) to the commutator norm (retained as a record; the entrywise-to-op
The Kostant Dirac D_G is block-diagonal in the Peter–Weyl decomposition, with eigenvalue $\pm D(\pi)$ on the isotypic component of π. For each isotypic component of f at weight σ, the multiplier matrix M_σ couples π' to π if and only if $\sigma \subset \pi \otimes \pi'^*$ in the Brauer–Klimyk decomposition. The commutator entry is

$$\big[D_G, M_\sigma\big]_{\pi,\pi'} \;=\; (D(\pi) - D(\pi')) \cdot (M_\sigma)_{\pi,\pi'}, \tag{31}$$

so by the Dirac-triangle inequality (20),

$$\|[D_G,\, M_\sigma]\|_{\mathrm{op}} \;\le\; \sqrt{C(\sigma)} \cdot \|M_\sigma\|_{\mathrm{op}}. \tag{32}$$

[3] Specifically, on the tested panels the analytical reduction to (SI′) is exhaustively case-checked by the numerical panel; at higher ranks (e.g., F_4, E_6, E_7, E_8) the analytical proof structure of Steps 1–5 is expected to carry through, but the per-orbit case-bookkeeping of Remark 3.19 would need to be re-tabulated on the relevant panels.

The bi-invariant Lipschitz norm of a unit-normalised harmonic Y_σ on G scales as $\|\nabla Y_\sigma\|_{L^\infty} \sim \sqrt{C(\sigma)}$ via the standard Sobolev embedding $H^1 \subset L^\infty$ at the level of fixed harmonic degree (this is the Laplace–Beltrami spectrum bound: $-\Delta Y_\sigma = C(\sigma) Y_\sigma$ implies $\|\nabla Y_\sigma\|_{L^2}^2 = C(\sigma)\|Y_\sigma\|_{L^2}^2$, with the L^∞ embedding bounded above by 1 at low harmonic level[4]). Combining (32) with the gradient scaling gives the per-isotypic Lipschitz comparison ratio

$$\frac{\|[D_G, M_\sigma]\|_{\mathrm{op}}}{\|\nabla Y_\sigma\|_{L^\infty}} \;\le\; C_3(G, \sigma), \tag{33}$$

with $C_3(G, \sigma) \le 1$ [retracted 2026-10-02: c3-per-summand-bridge] asymptotic-tight as $C(\sigma) \to \infty$ (the asymptotic-tightness verified numerically as $\rho_n = 1 - O(1/n)$ in the family $\lambda_n = (n, 0, \ldots, 0)$; see Proposition 3.14). Taking the supremum over σ gives the global bound (19) with $C_3(G) = \sup_\sigma C_3(G, \sigma) \le 1$, and the extension to $C^\infty(G)$ follows by density. □

Remark 3.23 (Transition cutoff Λ_* for tight asymptotic behaviour)**.** Lemma 3.11 gives the comparison with constant 1 at every cutoff, sharp as $\Lambda \to \infty$. (An earlier version stated finite-cutoff per-isotypic ratios $C_3(G, \sigma) < 1$, with the SU(2) closed form $\sqrt{(2j-1)/(2j+1)}$, i.e. 0.91 at $j = 5$ and 0.995 at $j = 50$; under $N = 2j$ that is the per-harmonic ratio Paper 38 withdrew as false on 2026-09-28, and it is withdrawn here.) How fast the compressed commutator norm of a fixed f approaches its limit as Λ grows is open and not required for the state-space GH convergence theorem.

Remark 3.24 (Recovery of Paper 38)**.** For $G = \mathrm{SU}(2)$, the Brauer–Klimyk decomposition reduces to the classical Clebsch–Gordan rule $V_j \otimes V_{j'} = \bigoplus_{|j-j'| \le k \le j+j'} V_k$, and the Dirac-triangle inequality (20) becomes $|j - j'| \le \sqrt{k(k+1)}$, which holds trivially because $k \ge |j - j'|$ implies $k(k+1) \ge k^2 \ge |j - j'|^2$. The intermediate inequality (21) is $|j - j'|^2 \le k(k+1)$ in the SU(2) case, again trivial. (An earlier version added that, combined with the Avery 3-Y selection rule, this recovers a per-harmonic ratio $\sqrt{(N-1)/(N+1)}$ of Paper 38; that per-harmonic inequality is false and was withdrawn in Paper 38, 2026-09-28.)

3.4 Lemma L4: Berezin reconstruction

Definition 3.25 (Berezin reconstruction map)**.** For $\Lambda > 0$ and $f \in C^\infty(G)$, define

$$B_\Lambda(f) \;:=\; P_\Lambda \cdot M_{K_\Lambda * f} \cdot P_\Lambda, \tag{34}$$

where K_Λ is the central spectral Fejér kernel of Definition 3.3 and the convolution is taken with respect to the normalised bi-invariant Haar measure on G. The convolution form is the definition; its Peter–Weyl expansion carries the multiplier-symbol weights $m_{K_\Lambda}(\pi)$ of Lemma 3.4(b), supported on the doubled Clebsch–Gordan window — *not* the mass-distribution weights $\dim V_\pi / Z_\Lambda$ (Remark 3.5; the two forms disagree already at $f = 1$). All four properties of Lemma 3.26 below are proved from the convolution form.

Lemma 3.26 (L4: properties of the Berezin reconstruction)**.** *The map $B_\Lambda : C^\infty(G) \to \mathcal{O}_\Lambda$ satisfies:*

(a) (Positivity.) If $f \ge 0$ on G, then $B_\Lambda(f)$ is positive semi-definite as an operator on $\mathcal{H}_\Lambda$.

(b) (Contractivity.) $\|B_\Lambda(f)\|_{\mathrm{op}} \le \|f\|_{L^\infty}$.

(c) (Approximate identity.) $\|B_\Lambda(f) - P_\Lambda M_f P_\Lambda\|_{\mathrm{op}} \le \gamma_\Lambda(G) \cdot \|\nabla f\|_{L^\infty}$.

(d) (L3 compatibility.) $\|[D_G, B_\Lambda(f)]\|_{\mathrm{op}} \le C_3(G) \cdot \|\nabla f\|_{L^\infty} \le \|\nabla f\|_{L^\infty}$.

[4] For higher harmonic levels the Sobolev embedding constant grows polynomially, but this is absorbed into the relative-multiplier-norm factor that defines $C_3(G)$ as the asymptotic supremum of the per-isotypic ratio; see the SU(2) case in [26, §3.3] for the precise bookkeeping.

Proof. **(a)** The convolution $g := K_\Lambda * f$ is non-negative on G as the convolution of two non-negative functions (Lemma 3.4(a)). Compression by an orthogonal projection preserves positivity: for any $\psi \in \mathcal{H}_\Lambda$,

$$\langle \psi | P_\Lambda M_g P_\Lambda | \psi \rangle \;=\; \int_G g(\omega)\, |P_\Lambda \psi(\omega)|^2 \, d\Omega \;\geq\; 0.$$

(b) By Young's inequality on the compact group G, $\|K_\Lambda * f\|_{L^\infty} \leq \|K_\Lambda\|_{L^1} \|f\|_{L^\infty} = \|f\|_{L^\infty}$. Compression contracts operator norms.

(c) The standard Lipschitz/good-kernel estimate gives

$$\|K_\Lambda * f - f\|_{L^\infty} \;\leq\; \int_G K_\Lambda(g)\, |f(g \cdot x) - f(x)|\, dg \;\leq\; \gamma_\Lambda(G)\, \|\nabla f\|_{L^\infty}.$$

Compression then gives the operator-norm bound.

(d) The cleanest argument is via the convolution form. Let $g := K_\Lambda * f \in C^\infty(G)$. Since $K_\Lambda \geq 0$ and $\|K_\Lambda\|_{L^1} = 1$, Young's inequality for the gradient, together with the fact that left-translation by an element of G commutes with the gradient on the bi-invariant metric, gives

$$\|\nabla g\|_{L^\infty} \;=\; \|\nabla (K_\Lambda * f)\|_{L^\infty} \;=\; \|K_\Lambda * \nabla f\|_{L^\infty} \;\leq\; \|\nabla f\|_{L^\infty}. \tag{35}$$

Applying Lemma 3.11 to g in the truncated multiplier representation gives

$$\|[D_G, B_\Lambda(f)]\|_{\mathrm{op}} \;\leq\; \|[D_G, M_g]\|_{\mathrm{op}} \;\leq\; C_3(G) \cdot \|\nabla g\|_{L^\infty} \;\leq\; C_3(G) \cdot \|\nabla f\|_{L^\infty} \;\leq\; \|\nabla f\|_{L^\infty},$$

where the final inequality uses $C_3(G) = 1$ from Lemma 3.11. □

Remark 3.27 (Non-Kähler analog at all ranks)**.** The Berezin reconstruction (34) is the non-Kähler central-Fejér-kernel analog of the Hawkins construction [11]. For general compact G, G is parallelisable but not Kähler unless G is a torus, so the Hawkins-style holomorphic Toeplitz form does not apply. The central-spectral-Fejér construction works at all ranks via Peter–Weyl + the central conjugacy-class measure, and is the natural generalisation of the S^1 Fejér reconstruction of van Suijlekom [8] and the spectral Fejér kernel of Leimbach–van Suijlekom [21] to non-abelian compact groups.

3.5 Lemma L5: state-space GH assembly via the approximation pair

We bound van Suijlekom's state-space Gromov–Hausdorff distance $d_{\mathrm{GH}}(\mathcal{T}_\Lambda, \mathcal{T}_G)$ (§2.5) with an approximation pair (B_Λ, P_Λ), $B_\Lambda : \mathcal{T}_G \to \mathcal{T}_\Lambda$ a unital completely positive (UCP) map and P_Λ the spectral compression, exactly as Paper 38's Lemma L5 does for SU(2). For orientation only: in Latrémolière's framework the same pair would be the data of a tunnel τ between the two triples, and the propinquity would be

$$\inf_\tau \mathrm{length}(\tau), \qquad \mathrm{length}(\tau) = \max(\mathrm{reach}(\tau), \mathrm{height}(\tau)); \tag{36}$$

that quantity is *not* what is proved here (the dual-reach hypothesis is the named gap). The four tunnel terms are recorded below for what a Latrémolière tunnel would require; the state-space GH bound itself is *not* obtained from this pair (see the proof), but from the lifted-state route of Remark 3.5.

Lemma 3.28 (L5: state-space GH bound from the approximation pair)**.** *The pair (B_Λ, P_Λ) with B_Λ from Definition 3.25 and P_Λ the spectral projection of (8) is an approximation pair between $\mathcal{T}_G$ and $\mathcal{T}_\Lambda$. The state-space GH bound*

$$d_{\mathrm{GH}}(\mathcal{T}_\Lambda, \mathcal{T}_G) \;\leq\; \gamma_\Lambda(G), \tag{37}$$

with $\gamma_\Lambda(G)$ from Lemma 3.4(d), holds unconditionally for $G = \mathrm{SU}(2)$ (Paper 38, `thm:main_unconditional`) and, for general G, conditionally on the per-group lifted-state decomposition of Remark 3.5. The pair does not supply an independent proof *(corrected 2026-09-04): of its four tunnel constituents,* $\mathrm{reach}_B \le \gamma_\Lambda(G)$ *is proved (Lemma 3.26(c)),* reach_P *is a named gap, and both height constituents are refuted ((39)).*

Proof. We record the four quantities a Latrémolière tunnel would carry — reach (forward and dual) and height (forward and dual). Paper 38's Lemma L5 organised its assembly (since withdrawn) by the two forward ones; what survives of that lemma is the record below, not a proof.

Reach. By Lemma 3.26(c), the forward reach satisfies $\mathrm{reach}_B \le \gamma_\Lambda(G) \cdot \|\nabla f\|_{L^\infty} \le \gamma_\Lambda(G)$ on the unit Lipschitz ball $\{\|f\|_{\mathrm{Lip}} \le 1\}$, where $\|\nabla f\|_{L^\infty} = \|f\|_{\mathrm{Lip}}$ by Connes' formula (step (b) of the proof of Lemma 3.11). The dual-reach estimate $\mathrm{reach}_P \le \gamma_\Lambda(G)$ is a *named gap* (Remark 3.5): the smoothing map has cb-norm 1, and a forward cb-norm controls no partial inverse. A corrected proof requires an antiderivative-transference lemma in the style of Leimbach–van Suijlekom [21]; for the scalar sector the conclusion is corroborated at the state-space Gromov–Hausdorff level for all compact metric groups by [10].

Height. The height-type quantity (Latrémolière's height for metric spectral triples [18]) measures the Lipschitz-distortion of the pair's maps with respect to the Lipschitz seminorms,

$$\mathrm{height}_B \;:=\; \sup_{\|f\|_{\mathrm{Lip}} \le 1} \big|\, \|f\|_{\mathrm{Lip}} - \|B_\Lambda(f)\|_{\mathrm{Lip}}^{\mathcal{O}_\Lambda} \big|. \tag{38}$$

Withdrawn 2026-09-04 [retracted 2026-09-04: l5-height-bound-achieved]. The text printed here until that date concluded $\mathrm{height}_B \le \gamma_\Lambda(G)$ (from the good-kernel estimate) and $\mathrm{height}_P = 0$ (from orthogonality of P_Λ). Both are false, at *every* rank, by the witness Paper 38 [26] records for the rank-one case, whose argument is independent of G: B_Λ is a finite-band reconstruction, so the unit-Lipschitz ball contains an f with $B_\Lambda(f) = 0$, for which the quantity in (38) equals $\|f\|_{\mathrm{Lip}} = 1$. Hence

$$\mathrm{height}_B \;\equiv\; 1 \quad \text{at every cutoff and every rank,} \tag{39}$$

and $\mathrm{height}_B \le \gamma_\Lambda(G)$ [retracted 2026-09-04: l5-height-bound-achieved] fails whenever $\gamma_\Lambda(G) < 1$; the same witness gives $\mathrm{height}_P \ne 0$.

The conclusion (37) *is unaffected*, for the reason it is unaffected in Paper 38: the state-space GH assembly bounds d_{GH} using reach-type estimates alone, with no height constituent. What is withdrawn is the height leg of the four-term organisation above — retained because it names what a Latrémolière tunnel would additionally require — and not the bound. Whether a *height-controlled* assembly exists for this pair at general rank is open, and Theorem 4.1 must not be cited as height-controlled.

Assembly. Until 2026-09-04 this proof closed with $d_{\mathrm{GH}} \le \max(\gamma, \gamma, \gamma, 0) = \gamma$ [retracted 2026-09-04: l5-height-bound-achieved], whose third and fourth arguments are the two refuted height constituents, and declared "all other constituents proved above". Withdrawn: with the heights refuted and reach_P a named gap, the pair establishes nothing toward (37) on its own. The bound holds by the lifted-state route — Paper 38's `thm:main_unconditional` at $G = \mathrm{SU}(2)$, which uses neither B_Λ nor any dual reach, and its per-group extension of Remark 3.5 at general G, conditional on the spin-window decomposition there — which is precisely the scope Theorem 4.1 states. Whether a height-controlled or dual-reach-controlled assembly of this pair exists is open. □

4 Main theorem and corollaries

Theorem 4.1 (Unified GH-convergence on compact semisimple Lie groups)**.** *Let G be a compact connected semisimple Lie group of rank $r \ge 1$ with bi-invariant Riemannian metric in the dual-*

Coxeter normalisation, and let $\mathcal{T}_G$ be its canonical metric spectral triple. Let $\mathcal{T}_\Lambda$ be its Connes–van Suijlekom truncation at Casimir cutoff $\Lambda > 0$. Then $\mathcal{T}_\Lambda$ converges to $\mathcal{T}_G$ in van Suijlekom's state-space Gromov–Hausdorff distance, with the state spaces metrized by the left-translation Lipschitz seminorm (unconditionally for SU(2)*; for general G conditional on the per-group spin-window lifted-state decomposition of Remark 3.5):*

$$d_{\mathrm{GH}}(\mathcal{T}_\Lambda, \mathcal{T}_G) \;\le\; \gamma_\Lambda(G) \;\to\; 0, \qquad \gamma_\Lambda(\mathrm{SU}(2)) = \frac{2}{\pi}\,\frac{\log\Lambda}{\Lambda} + O\big(1/\Lambda\big), \tag{40}$$

where:

(i) The bound has constant 1 *(the lifted-state route); the Lipschitz comparison constant $C_3(G) = 1$ of Lemma 3.11 (by compression, sharp as $\Lambda \to \infty$) enters only the withdrawn approximation-pair route (corrected 2026-10-02: an earlier display read $C_3(G)\cdot\gamma_\Lambda(G)$, which the lifted-state route does not supply).*

(ii) For $G = \mathrm{SU}(2)$ the displayed asymptotic holds ***[INTERNAL THEOREM]*** *(Paper 38's $(4/\pi)\log n/n$ at $n \simeq 2\Lambda$). For the other simple G the bound holds at the same tier with $\gamma_\Lambda(G) \to 0$; the rate form, and whether the leading Fejér-moment constant is the same, independent of the rank, $|W(G)|$ and the Lie type, are* ***[OPEN]*** *(proposed mechanism: Remark 3.6; the numerical extractions of Remark 3.7 compare mismatched conventions; status note in §3.2). For semisimple G with more than one simple factor the bound holds at the same tier with $\gamma_\Lambda(G) \to 0$, but the leading constant and the rate are open (Remark 3.9).*

Proof. Lemma 3.1 supplies the operator-system substrate; Lemma 3.4 the central spectral Fejér kernel and its moment, with $\gamma_\Lambda(\mathrm{SU}(2)) \sim (2/\pi)\log\Lambda/\Lambda$; Lemma 3.11 (used only by the withdrawn route) the Lipschitz comparison constant $C_3(G) = 1$, proved by compression at every rank (the Dirac-triangle analysis, Proposition 3.14, is a numerical panel of a representation-theoretic inequality, not a check of it); Lemma 3.26 the Berezin reconstruction with the four properties a Latrémolière tunneling pair would require; and—in the *withdrawn* route only—Lemma 3.28 the assembly into the state-space GH bound.

Which route establishes this. Not the L5 assembly: its dual reach is a named gap and both height constituents are refuted (§3.5, corrected 2026-09-04), so the approximation pair supplies no independent proof. The bound (37) holds on the lifted-state route inherited from Paper 38 — UCP compression plus the conjugation average, using no Berezin map, no partial inverse and hence no height term — and combining it with the asymptotic rate (16) (SU(2)) or with $\gamma_\Lambda(G) \to 0$ (Remark 3.9, every other G) gives (40). □

Proposition 4.2 (Limit identification)**.** *The state-space Gromov–Hausdorff limit of the sequence $(\mathcal{T}_\Lambda)_{\Lambda>0}$ (in van Suijlekom's distance; the strictly stronger Latrémolière propinquity limit is not claimed) is the round-G canonical metric spectral triple $\mathcal{T}_G$, with the Connes distance on $\mathcal{T}_G$ coinciding (by Kantorovich–Rubinstein duality) with the Wasserstein–Kantorovich distance with respect to the bi-invariant geodesic distance on G.*

Proof sketch. Theorem 4.1 guarantees $d_{\mathrm{GH}}(\mathcal{T}_\Lambda, \mathcal{T}_G) \to 0$, so the limit object exists in the state-space GH completion and equals $\mathcal{T}_G$. The L4(c) approximate-identity property pins the limit to be the actual triple (rather than an isomorphism class). □

We list explicit corollaries for the classical compact Lie groups, all of which are immediate consequences of Theorem 4.1.

Corollary 4.3 (SU(N))**.** *For $G = \mathrm{SU}(N)$ with the bi-invariant metric in the dual-Coxeter normalisation ($C(\mathrm{ad}) = N$), the truncated metric spectral triples $\mathcal{T}_\Lambda$ converge to $\mathcal{T}_{\mathrm{SU}(N)}$ in van Suijlekom's state-space Gromov–Hausdorff distance with the bound $\gamma_\Lambda(\mathrm{SU}(N)) \to 0$* at the tier of Theorem 4.1*: at $N = 2$, $\gamma_\Lambda = (2/\pi)\log\Lambda/\Lambda + O(1/\Lambda)$* ***[INTERNAL THEOREM]****; for $N \ge 3$ the leading constant is* ***[OPEN]*** *and the convergence is conditional on the per-group spin-window decomposition verified there for* SU(2)*.*

Corollary 4.4 (Spin(n)). *For $G = \mathrm{Spin}(n)$ with $n \geq 5$ (simple; $\mathrm{Spin}(3) = \mathrm{SU}(2)$ is Corollary 4.7, and $\mathrm{Spin}(4) = \mathrm{SU}(2) \times \mathrm{SU}(2)$ has two simple factors, so its constant is open) and the bi-invariant metric in the dual-Coxeter normalisation ($C(\mathrm{ad}) = n - 2$), the truncated triples converge with the Fejér-moment bound $\gamma_\Lambda(G) \to 0$, at the tier of Theorem 4.1 (conditional on the per-group spin-window decomposition; the leading constant **[OPEN]**).*

Corollary 4.5 (Sp(n)). *For $G = \mathrm{Sp}(n)$ with the bi-invariant metric in the dual-Coxeter normalisation ($C(\mathrm{ad}) = n+1$), the truncated triples converge with the Fejér-moment bound $\gamma_\Lambda(G) \to 0$, at the tier of Theorem 4.1 (conditional on the per-group spin-window decomposition; the leading constant **[OPEN]**).*

Corollary 4.6 (Exceptional groups). *For $G \in \{G_2, F_4, E_6, E_7, E_8\}$ with the bi-invariant metric in the dual-Coxeter normalisation, the truncated triples converge with the Fejér-moment bound $\gamma_\Lambda(G) \to 0$, at the tier of Theorem 4.1 (conditional on the per-group spin-window decomposition; the leading constant **[OPEN]**).*

Corollary 4.7 (Recovery of Paper 38). *For $G = \mathrm{SU}(2)$ (rank 1), Theorem 4.1 recovers Paper 38's unconditional theorem [26], in the same translation-seminorm metrization, with the same explicit bound and the same rate: Paper 38's $(4/\pi)\log n/n$, $4/\pi = \mathrm{Vol}(S^2)/\pi^2$, in its variable $n \simeq 2\Lambda$, which is $(2/\pi)\log\Lambda/\Lambda$ here.*

The corollaries are all immediate consequences of Theorem 4.1; the only G-dependent input is the choice of dual-Coxeter normalisation in the bi-invariant Killing form, which is fixed in advance by the declared rule (2) (not the field-standard normalisation; §2.1).

5 Is the Fejér-moment constant group-independent? An open question

Theorem 4.1(ii) leaves open whether the leading Fejér-moment constant $c(G)$ is the same on every simple group. In this section we discuss the two natural readings, why the numerical discriminator test reported here does not decide between them, the proposed cancellation mechanism, and the structural reading of the SU(2) constant as a master Mellin engine signature.

5.1 The two natural readings, and what discriminates them

Paper 38 identified the SU(2) constant, per its shell index n, as

$$c_n(\mathrm{SU}(2)) \;=\; 4/\pi \;=\; \mathrm{Vol}(S^2)/\pi^2 \;=\; 2\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)), \tag{41}$$

where the last expression is a quotient of unit-radius sphere volumes; in the cutoff Λ ($n \simeq 2\Lambda$), $c(\mathrm{SU}(2)) = 2/\pi$. (Paper 38 originally read this as the Hopf fibration's base-to-total ratio; that label was withdrawn on 2026-09-03 — the Hopf base is $S^2(r/2)$ and its ratio is $1/(2\pi)$ — without affecting the identity above.) Two natural rank-r generalisations were proposed in the supporting scoping memo [28], both normalised to the per-n value $4/\pi$:

Reading A (group-independent). $c(G) = c(\mathrm{SU}(2))$ for every compact connected simple Lie group G, regardless of rank or $|W|$.

Reading B (Weyl-formula). The constant scales with the Weyl group order and the rank: $c(G) = 2|W(G)|/\pi^r$, where $r = \mathrm{rank}(G)$.

Readings A and B agree at SU(2) ($|W| = 2, r = 1$: both give the per-n value $4/\pi$, not the per-Λ value $2/\pi$, so both would need restating in Λ). As formulated, they diverge substantially at higher rank:

G	$\lvert W\rvert$	r	A (per-n normalisation)	B: $2\lvert W\rvert/\pi^r$
SU(2)	2	1	1.273	1.273 (agrees)
SU(3)	6	2	1.273	1.216
Sp(2)	8	2	1.273	1.621
G_2	12	2	1.273	2.432

The G_2 predictions differ by 91%. The pipeline outputs of Remark 3.7 (Sp(2): 1.087, G_2: 1.177) sit closer to Reading A's column, but the pipeline returns $(2/\pi)/\sqrt{2} \approx 0.450$ on SU(2), not the anchor, and the columns are per-n values while the extraction is per Λ; the comparison is between mismatched conventions and decides nothing, at G_2 included. The discrimination is **[OPEN]**.

5.2 The proposed mechanism: the Plancherel × Vandermonde cancellation

If the leading constant is group-independent, the following cancellation (Remark 3.6 above) is the proposed mechanism — derived at rank 1 (Paper 38 Appendix A) and given as a proof sketch at rank ≥ 2, where the numerical extractions do not test it (the fully rank-uniform write-up is a named gap).

The central spectral Fejér kernel is built with Plancherel weights $\sqrt{\dim V_\pi}$ by definition of the central Fejér kernel construction (Definition 3.3). When the mass-concentration moment γ_Λ is computed via the Weyl integration formula (4), the Plancherel weights appear *squared* as $\dim V_\pi$ (one factor from each copy of the kernel in $|\cdot|^2$), and the Weyl integration Jacobian $|\Delta(t)|^2 = \prod_{\alpha>0} |2\sin(\langle\alpha, t\rangle/2)|^2$ appears as a multiplicative factor on the Cartan torus $T \subset G$. The Weyl dimension formula states

$$\dim V_\pi \;=\; \prod_{\alpha>0} \frac{\langle \lambda_\pi + \rho,\, \alpha\rangle}{\langle \rho,\, \alpha\rangle}, \tag{42}$$

which has the same Vandermonde-style $\prod_{\alpha>0}$ product structure as $|\Delta(t)|^2$. The sketch proposes that these two products cancel at leading order, which would eliminate all rank-, $|W|$-, and N_+-dependence (where N_+ is the number of positive roots); this is unproved.

What is left is the rank-1 (SU(2)) constant ($4/\pi$ per n, $2/\pi$ per Λ at rank 1) from each positive-root direction's Euler–Catalan asymptote

$$\sum_{d \text{ odd}} \frac{1}{d^2} \;=\; \frac{\pi^2}{8},$$

which is the same identity that gives the $4/\pi$ constant in Paper 38's rank-1 sum-rule calculation for SU(2) [26, App. A]. The rank-1 asymptotic

$$\gamma_n^{\mathrm{SU}(2)} \;\sim\; \frac{4}{\pi} \cdot \frac{\log n}{n}$$

is then applied to each of the N_+ positive-root directions, with the Plancherel × Vandermonde cancellation proposed to make the proportionality constant aggregate to the rank-1 value rather than to a rank-dependent product; the value this gives in Λ at rank ≥ 2 is unresolved (Remark 3.6). (On the circle comparison: per Paper 38's shell index n and in the dual-Coxeter normalisation the SU(2) constant is twice the classical circle Fejér constant, while on the unit S^3 the two coincide at $2/\pi$, the factor 2 being the metric scale (Paper 38, Remark "Connection to the circle Fejér estimate").)

Structural reading. If the proposed cancellation holds, the central spectral Fejér kernel would see the rank-1 (SU(2)) asymptotic analysis through Peter–Weyl, *not* through the global geometry of G/T. Under the proposal, the Plancherel weight $\sqrt{\dim V_\pi}$ and the Vandermonde Jacobian $|\Delta(t)|^2$, both with $\prod_{\alpha>0}$ structure tied to the same root system, would cancel at leading order in the ratio $\dim V_\pi^2/|\Delta(t)|^2$ that controls the dominant contribution to γ_Λ.

This cancellation, if it holds at rank ≥ 2, would give Reading A (group-independent $c(G)$). Reading B (Weyl-formula $c(G) = 2|W(G)|/\pi^r$) would require the Weyl-group order to imprint multiplicatively on the leading-order rate, which the cancellation mechanism excludes at sketch level; the numerical extractions do not discriminate (§5.1, Remark 3.7).

5.3 Identification with the master Mellin engine

[OBSERVATION] Paper 38's per-n constant $4/\pi$ is read (a reading, not a derivation; Paper 38) as the content of one specific sub-mechanism of a master Mellin engine on the spectral triple, in the following sense. The spectral-action heat kernel admits a Mellin transform

$$\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})](s) \;=\; \int_0^\infty t^{s-1}\mathrm{Tr}(D^k e^{-tD^2})dt, \tag{43}$$

where $k \in \{0, 1, 2\}$ indexes three structurally distinct sub-mechanisms by which transcendentals (specifically powers of π and Catalan-type Dirichlet L-values) enter spectral observables on the triple. The $k = 0$ sub-mechanism corresponds to the heat-kernel-as-volume-factor direction: here the leading transcendental signature on any compact homogeneous space is a volume factor, which for $G = \mathrm{SU}(2)$ is the unit-sphere volume quotient (per n: $4/\pi$ dual-Coxeter, $2/\pi$ on the unit S^3; per Λ: $2/\pi$ in every normalisation; Paper 38, volume-ratio note), not the Hopf fibration's base-to-total ratio.

Under that reading, a group-independent leading constant would be the rank-invariance of the $k = 0$ volume-slot value; that is **[OPEN]** (derived only at rank 1; §5.1). The GeoVac-internal scoping memo [28] originally proposed this identification as an open question; it remains a reading, and its rank-≥ 2 input is open.

6 Discussion and open problems

6.1 Quantitative tightness and the log-power test

Theorem 4.1 gives the SU(2) asymptotic constant $2/\pi$ in $\gamma_\Lambda(\mathrm{SU}(2)) \sim (2/\pi)\log\Lambda/\Lambda$ but not the next-order constant; at rank ≥ 2 the leading constant itself is open. Numerical fits at the panel size used in this work ($\Lambda^2 \in [1, 1000]$) display Stein–Weiss bias of 3–6% typical and 10–25% at smaller-asymptotic panels; a Reading A versus Reading B verdict would in addition need the extraction and the readings put in one convention (§5.1), which is not done here.

A separate open question is the *log-power*: at rank $r \geq 2$, is the rate single-log ($\log\Lambda/\Lambda$) or could it admit a double-log ($\log^2\Lambda/\Lambda$) refinement? This question is *numerically inconclusive* at the Λ ranges accessible to the verification panel: a $\log^r\Lambda/\Lambda$ ansatz with $r = 2$ fits the panel within the Stein–Weiss bias on SU(3), Sp(2), and G_2, and a clean discriminator would require $\Lambda^2 \gtrsim 10^5$ which is computationally expensive; at rank ≥ 2 the rate form, like the constant, is open. The structural implications differ: if the rate is single-log on every simple group, the M1 master-Mellin reading is sharpest; if it splits into a $\log^r$ family at rank r, each log-power level would correspond to a separate Cesaro-moment sub-tier.

6.2 Rank-3 numerical extraction and compute scaling

The proposed Plancherel × Vandermonde cancellation (Remark 3.6) supplies the *mechanism* proposed for a group-independent $c(G)$ on simple G — derived at rank 1, a proof sketch at

rank ≥ 2 (the fully rank-uniform write-up is a named gap). The numerical pipeline outputs (SU(3): 1.243, Sp(2): 1.087, G_2: 1.177) carry the convention mismatch of Remark 3.7 and do not test it. At SU(4) (rank 3) the output is 0.900, with the compute budget terminating the panel at $\Lambda^2 = 300$.

Two factors degrade the extraction at rank 3. First, the rank-3 Weyl integration is genuinely 3-dimensional, increasing the per-point quadrature cost by roughly an order of magnitude over rank 2; Cartan-torus rows at $\Lambda^2 \approx 400$ take 15–30 minutes each. Second, the pre-asymptotic structure is stronger at rank 3 than at rank 2, making the leading-order extraction more sensitive to the fit form. The forced-coefficient χ^2 comparison of Reading A with Reading B at SU(4) inherits the convention mismatch and is not a test. The SU(4) data add a third rank but, like the rank-2 data, do not discriminate between the readings until the pipeline's normalisation is corrected. Detailed extraction analysis is in the supporting memo [31].

6.3 Extension to compact symmetric spaces G/K

The natural next-step generalisation is from compact Lie groups (Class 1 in the scoping memo) to compact Riemannian symmetric spaces G/K (Class 2). The state-space GH convergence theorem at full Class 2 generality would subsume Theorem 4.1 as the "group case" $G \times G/G_{\mathrm{diag}}$ and would additionally cover spheres $S^n = \mathrm{SO}(n+1)/\mathrm{SO}(n)$ for $n \geq 2$, complex projective spaces $\mathbb{CP}^n = \mathrm{SU}(n+1)/(U(n) \times U(1))$, complex Grassmannians, and full flag varieties.

The technical overhead is the K-invariance bookkeeping in the Peter–Weyl decomposition $C^\infty(G/K) = \bigoplus_\pi V_\pi^K$ and the corresponding zonal-spherical-function machinery. The central spectral Fejér kernel construction extends to a K-bi-invariant class-function kernel on G (spherical functions of Helgason [14]), and the Lipschitz comparison transfers by compression wherever the Dirac operator commutes with the truncation. Whether the SU(2) moment constant persists at that level of generality is open, as it is for the simple groups; the verification at rank ≥ 2 symmetric spaces is deferred to a follow-on paper.

6.4 Relation to Rieffel's coadjoint-orbit programme

A complementary direction in the quantum metric convergence literature is Rieffel's coadjoint-orbit programme [22, 23], which constructs G-invariant Dirac operators on coadjoint orbits $\mathcal{O} \subset \mathfrak{g}^*$ of a compact Lie group G and proves that fuzzy approximations (matrix algebras of growing dimension) converge to $C(\mathcal{O})$ in spectral propinquity. This work covers compact symmetric spaces that arise as coadjoint orbits (notably $S^2 = \mathrm{SU}(2)/U(1)$, $\mathbb{CP}^n$, and complex Grassmannians) and converges *from below* (matrix algebras of growing dimension).

Theorem 4.1 converges *from above* (spectral truncation of the continuum object) and treats compact Lie groups G themselves (which are not coadjoint orbits but rather principal homogeneous spaces for G acting on itself by left translation). The two directions are categorically complementary: Rieffel's framework does not cover $S^3 = \mathrm{SU}(2)$, and Theorem 4.1 does not cover $S^2 = \mathrm{SU}(2)/U(1)$. The natural unified statement, which would subsume both, lives at the level of compact symmetric spaces G/K discussed in §6.3.

6.5 Lorentzian / Wick-rotated analog

The Kostant cubic Dirac spectral triple is Riemannian, with KO-dimension $\dim G \pmod 8$. The Lorentzian analog, via Wick rotation of G to a non-compact form, carries different KO-dimension and sign convention, and may admit a different convergence theory entirely. The Bizi–Brouder–Besnard framework [2] for $(m, n) \in (\mathbb{Z}/8)^2$ Krein-class spectral triples provides the algebraic backbone for the Lorentzian extension, but the corresponding quantum-Gromov–Hausdorff theory does not yet exist in the published literature; a Lorentzian Latrémolière-style propinquity for Krein-class triples remains open.

6.6 Cross-manifold tensor products and the Coulomb/HO asymmetry

A natural extension of Theorem 4.1 would aim at tensor products of spectral triples on different manifolds, in particular $\mathcal{T}_G \otimes \mathcal{T}_H$ for compact Lie groups G, H of possibly different rank. The same-manifold tensor product $\mathcal{T}^{\lambda_a}_{\mathrm{SU}(2)} \otimes \mathcal{T}^{\lambda_b}_{\mathrm{SU}(2)}$ at distinct focal lengths is the subject of Paper 39 [27]; the genuine cross-manifold case $\mathcal{T}_G \otimes \mathcal{T}_H$ with $G \neq H$ should extend from Paper 39 along its *lifted-state* route, which uses neither a Berezin map nor a partial inverse and so has no manifold-specific step to generalise. We have not written out the details. *(Corrected 2026-09-07: this paragraph previously proposed extending "with the Pythagorean Leibniz constant generalising to a Cartan-product-type bound". That refinement is* ***withdrawn*** *in Paper 39 — graded anti-commutation does not give operator-norm orthogonality — so the extension was aimed at an abandoned route. The joint constant of that route is now $C_3^{(2)} = 1$, by compression (Paper 39, Lemma L3-T), and is not a rate constant of Paper 39's theorem.)*

6.7 Connection to Hekkelman–McDonald non-commutative integral

Hekkelman and McDonald [13] construct a non-commutative integral on spectrally truncated triples and link it to a Szegö limit formula, for spectrally truncated spectral triples in general (tori are among their examples); it is a noncommutative-integral result, not a Gromov–Hausdorff convergence result. The underlying truncation framework is the Connes–van Suijlekom paradigm common with the present paper. We expect that the non-commutative integral of [13] admits a class-wide compact-Lie-group extension under the same Peter–Weyl machinery used here, with the Fejér-moment constant appearing as the leading Szegö-type asymptotic. The detailed generalisation is open and would constitute a Szegö companion to the present convergence theorem.

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References

[1] I. Agricola, *Connections on naturally reductive spaces, their Dirac operator and homogeneous models in string theory*, Comm. Math. Phys. **232** (2003), 535–563.

[2] N. Bizi, C. Brouder, and F. Besnard, *Space and time dimensions of algebras with applications to Lorentzian noncommutative geometry and quantum electrodynamics*, J. Math. Phys. **59** (2018), 062303; arXiv:1611.07062.

[3] M. Bożejko and G. Fendler, *Herz–Schur multipliers and uniformly bounded representations of discrete groups*, Arch. Math. **57** (1991), 290–298.

[4] N. Bourbaki, *Groupes et algèbres de Lie, Chapitres 7–9*, Eléments de mathématique, Hermann/Springer, 1975 (English translation: Springer, 2005).

[5] T. P. Branson, *Stein–Weiss operators and ellipticity*, J. Funct. Anal. **151** (1997), 334–383.

[6] A. Connes, *Noncommutative Geometry*, Academic Press, 1995.

[7] A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Comm. Math. Phys. **383** (2021), 2021–2067; arXiv:2004.14115.

[8] W. D. van Suijlekom, *Gromov–Hausdorff convergence of state spaces for spectral truncations*, J. Geom. Phys. **162** (2021), 104075; arXiv:2005.08544.

[9] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics **129**, Springer, 1991.

[10] Y. Gaudillot-Estrada and W. D. van Suijlekom, *Convergence of spectral truncations for compact metric groups*, Int. Math. Res. Not. (IMRN) **2025** (2025), rnaf197; arXiv:2310.14733.

[11] E. Hawkins, *Geometric quantization of vector bundles and the correspondence with deformation quantization*, Comm. Math. Phys. **215** (2000), 409–432.

[12] E. Hekkelman, *Truncated geometry on the circle*, Lett. Math. Phys. **112** (2022), 20; arXiv:2111.13865.

[13] E. Hekkelman and E. McDonald, *A noncommutative integral on spectrally truncated spectral triples, and a link with quantum ergodicity*, J. Funct. Anal. **289**(12) (2025) 111154; arXiv:2412.00628.

[14] S. Helgason, *Groups and Geometric Analysis*, Academic Press, 1984.

[15] V. G. Kac, *Infinite-Dimensional Lie Algebras*, 3rd ed., Cambridge Univ. Press, 1990.

[16] B. Kostant, *A cubic Dirac operator and the emergence of Euler number multiplets of representations for equal rank subgroups*, Duke Math. J. **100** (1999), 447–501.

[17] F. Latrémolière, *The dual Gromov–Hausdorff propinquity*, J. Math. Pures Appl. **103** (2015), no. 2, 303–351; arXiv:1311.0104.

[18] F. Latrémolière, *The Gromov–Hausdorff propinquity for metric spectral triples*, Adv. Math. **404** (2022), Paper No. 108393, 56pp; preprint arXiv:1811.10843, 2018.

[19] F. Latrémolière, *The quantum Gromov–Hausdorff propinquity*, Trans. Amer. Math. Soc. **368** (2016), 365–411; arXiv:1302.4058.

[20] M. Leimbach, *Convergence of Peter–Weyl truncations of compact quantum groups*, J. Noncommut. Geom., to appear (2024); arXiv:2409.16698.

[21] M. Leimbach and W. D. van Suijlekom, *Gromov–Hausdorff convergence of spectral truncations for tori*, Adv. Math. **439** (2024), Paper No. 109496.

[22] M. A. Rieffel, *Dirac operators for coadjoint orbits of compact Lie groups*, Münster J. Math. **2** (2009), 265–298; arXiv:0812.2884.

[23] M. A. Rieffel, *Dirac operators for matrix algebras converging to coadjoint orbits*, Comm. Math. Phys. **401** (2023), 1951–2008; arXiv:2108.01136.

[24] E. M. Stein and G. Weiss, *Introduction to Fourier Analysis on Euclidean Spaces*, Princeton Math. Ser. **32**, Princeton Univ. Press, 1971.

[25] A. Zygmund, *Trigonometric Series, Vol. I and II*, 3rd ed., Cambridge Univ. Press, 2002.

[26] J. Loutey, *State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3*, GeoVac internal preprint (May 2026), DOI via Zenodo.

[27] J. Loutey, *State-space Gromov–Hausdorff convergence of truncated tensor-product Camporesi–Higuchi spectral triples on $S^3 \times S^3$*, GeoVac internal preprint (May 2026), DOI via Zenodo.

[28] J. Loutey, *Unified GH-convergence scoping memo* (`debug/unified_gh_scoping_memo.md`), GeoVac internal working note (May 2026).

[29] J. Loutey, Sp(2) *and G_2 rate constant: A vs B discriminator test* (`debug/sp2_g2_rate_constant_memo.md`), GeoVac internal working note (May 2026).

[30] J. Loutey, *Dirac-triangle inequality: full proof with PRV-summand bound at all ranks* [retracted 2026-10-02: int-from-prv-theorem-b], GeoVac internal working note (May 2026); title as filed — its all-ranks claim is superseded, Theorem 3.16(b) falling short of (INT) by $|\rho|^2$.

[31] J. Loutey, *SU(4) rate constant extraction at rank* 3*: compute-budget analysis and forced-coefficient verdict*, GeoVac internal working note (May 2026).

[32] S. Kumar, Proof of the Parthasarathy–Ranga Rao–Varadarajan conjecture, *Invent. Math.* **93** (1988), 117–130.

[33] O. Mathieu, Construction d'un groupe de Kac–Moody et applications, *Compos. Math.* **69** (1989), 37–60.

[34] P. Polo, Variétés de Schubert et excellentes filtrations, *Astérisque* **173–174** (1989), 281–311.

[35] J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Graduate Texts in Mathematics **9**, Springer, New York, 1972 (the dominance partial order: §10.1, §13.2).